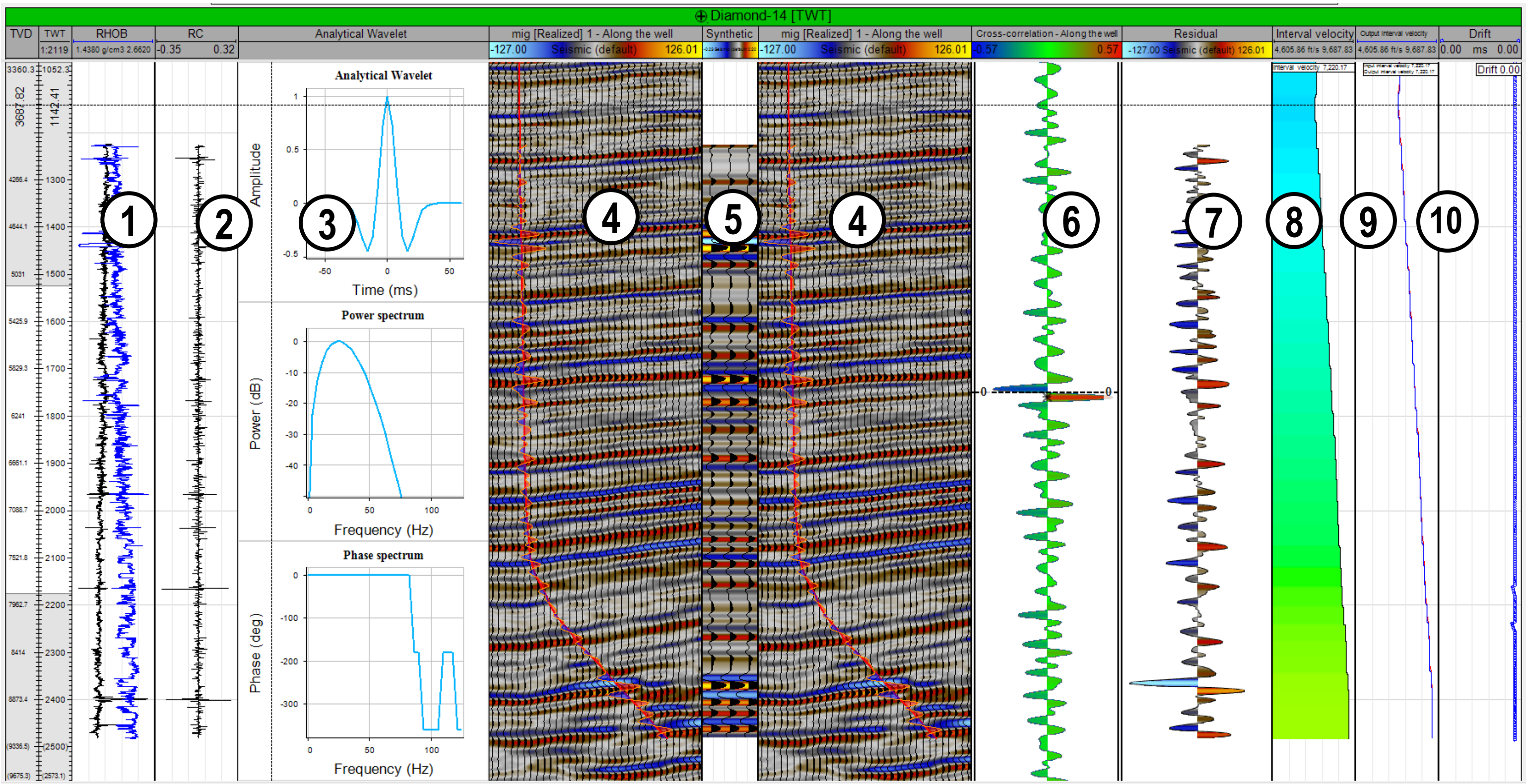
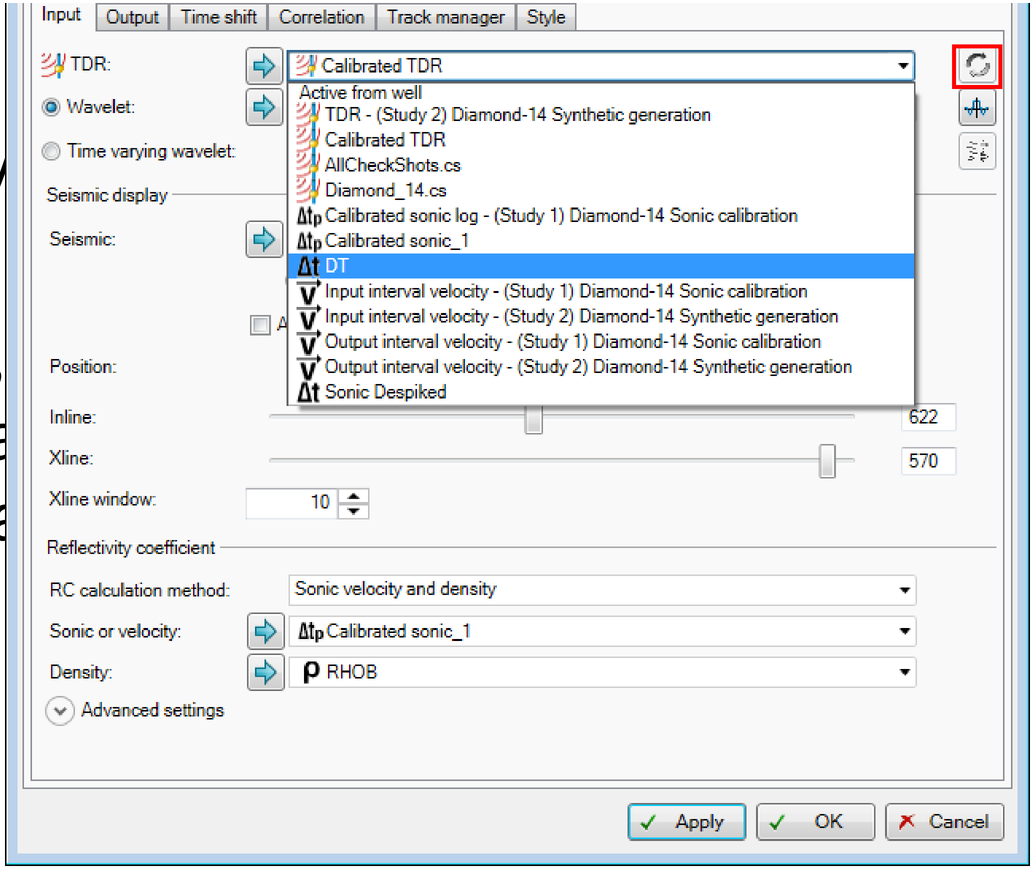
Generate a synthetic seismogram



1 Input logs track
2 Reflectivity track
3 Wavelet track
4 Seismic track
5 Synthetic track
6 Cross-correlation track
7 Residual track
8 Interval Velocity track
9 Input/Output Interval Velocity track
10 Drift track

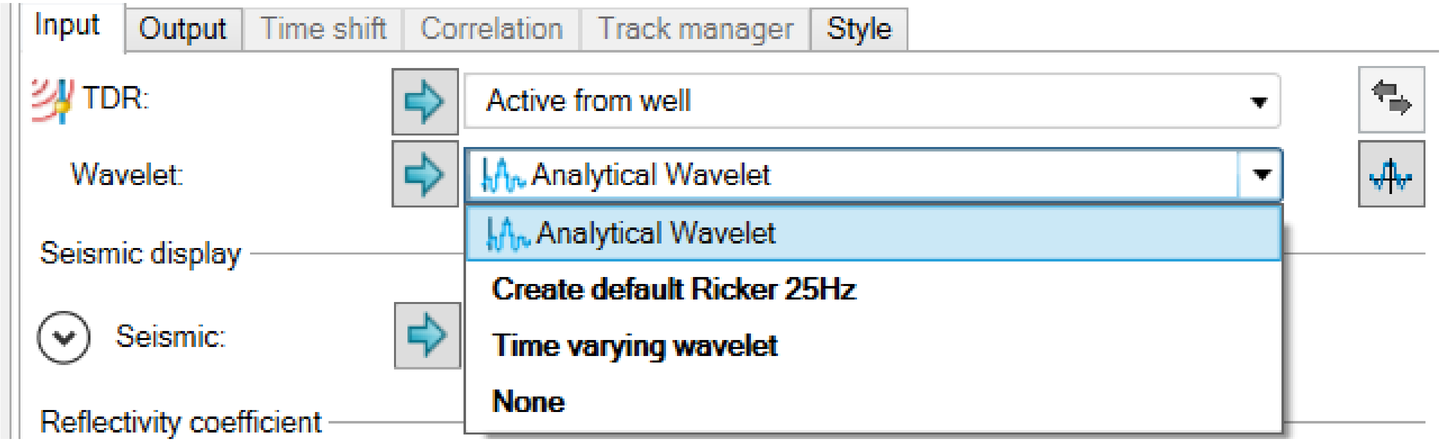
Select TDR input

- Seismic to well tie process is very interactive. You can select many ty of TDR as input in the study.
- You can select any TDR, sonic log, velocity log contained by the well a TDR input for the synthetics genera study.



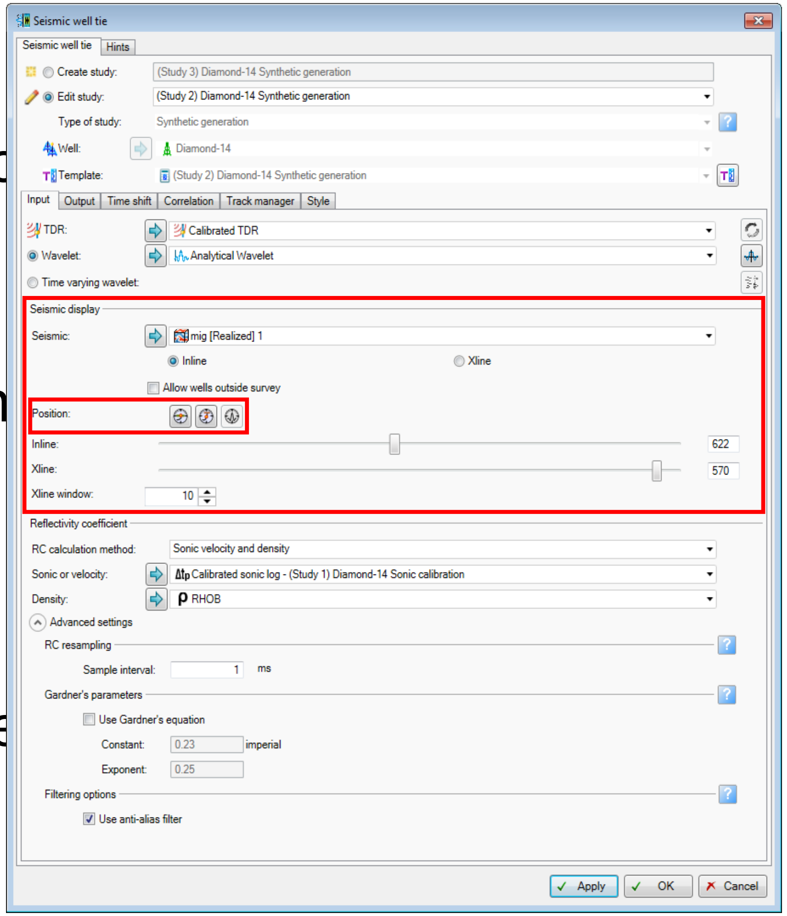
Wavelet

- Insert the wavelet to be convolved with the reflectivity series to create a synthetic seismogram.
- If no wavelet is available in the project, open the wavelet toolbox to create a new one by clicking *Lau*  *wavelet toolbox* .



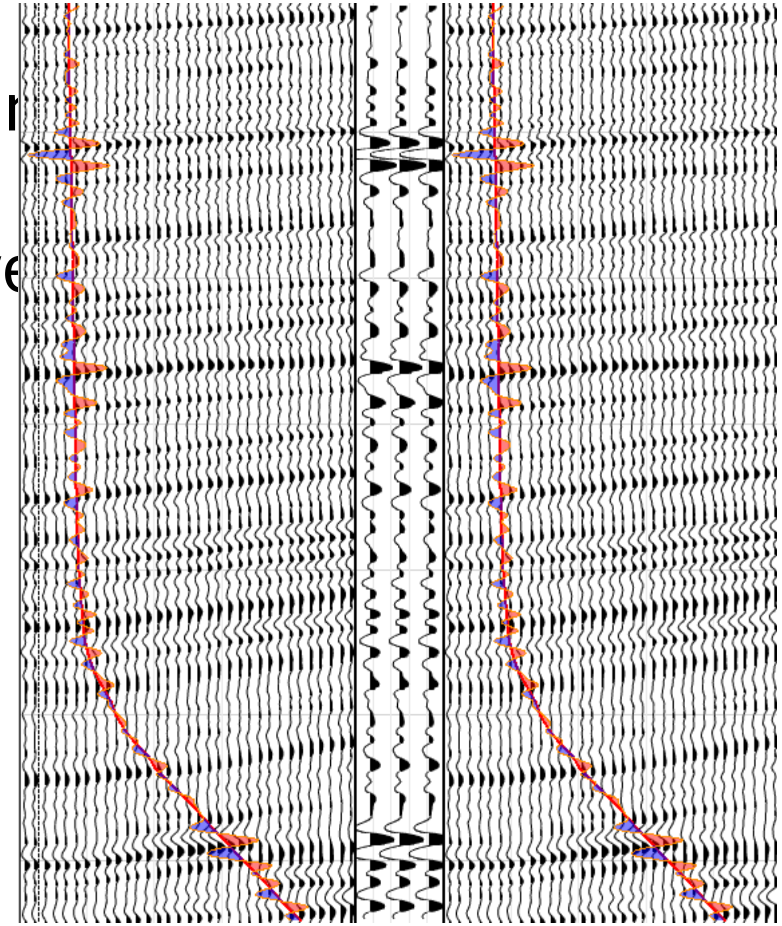
Seismic display

- In the **Seismic display** section, you inse seismic that is used as a reference to c the seismic-synthetic tie.
- You can use a 3D cube or a 2D line.
- There are four seismic display position methodologies for the Seismic well tie
  - Seismic along the well trajectory
  - Wellhead location on Inline/Xline
  - Deviated well location on Inline/Xline
  - Selected Wavelet on Inline/Xline



Seismic along the well trajectory option

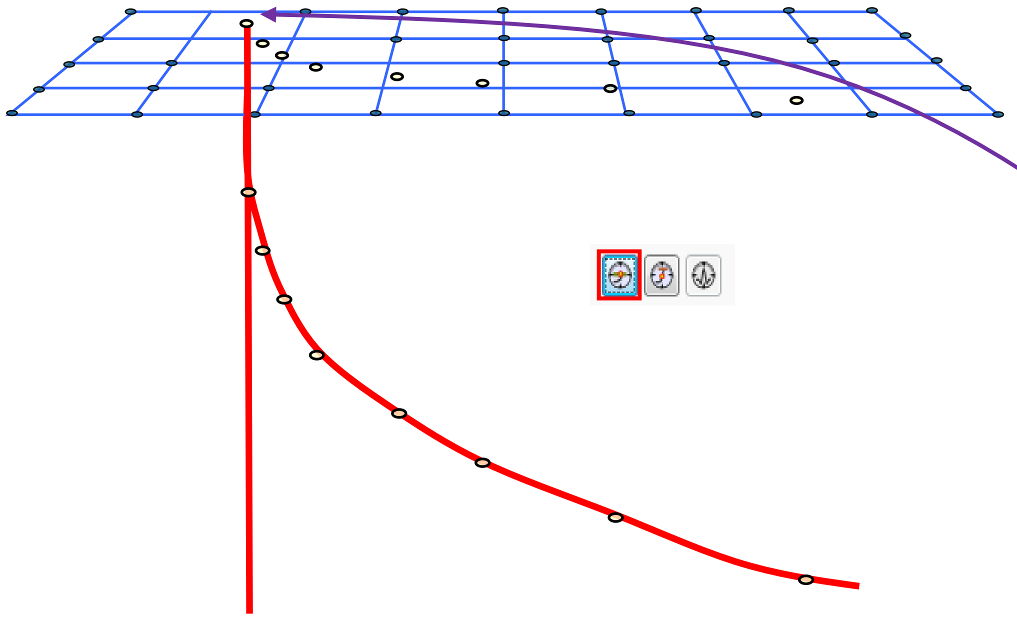
- This option is the default visualizatior
- The seismic is displayed along the we
  The option allows a deviated well.

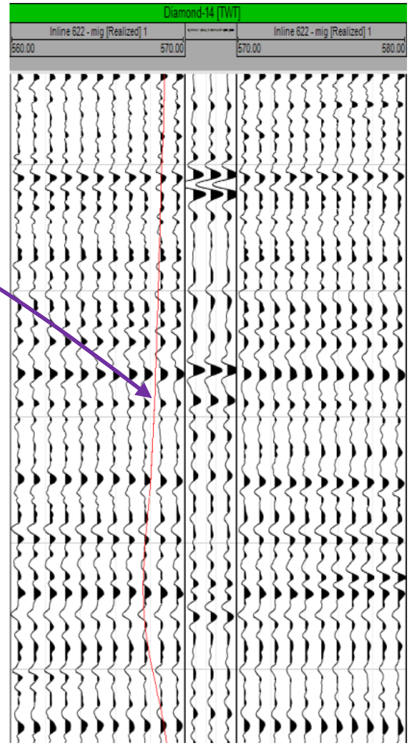


Wellhead location

- Set the Inline-Xline to the location of the wellhead.

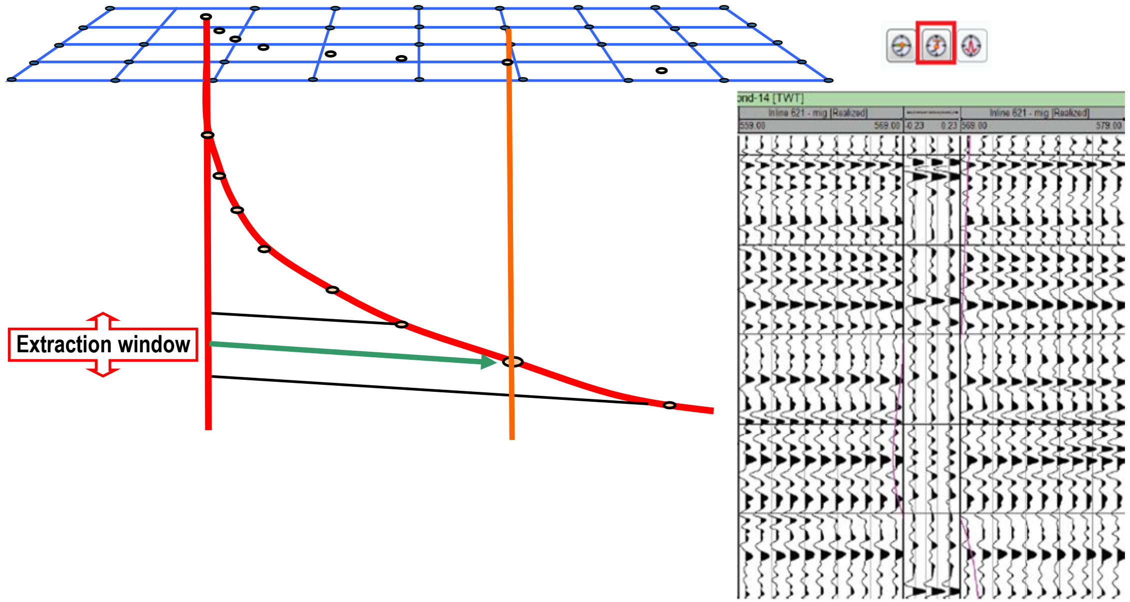
Position is set at the well surface XY location.





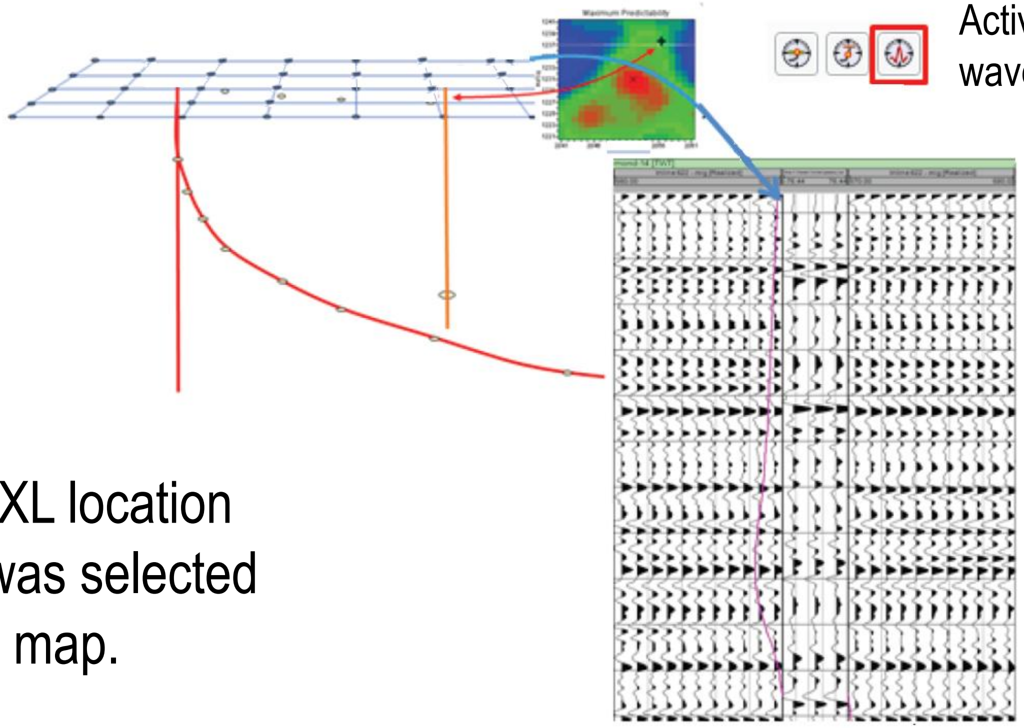
Deviated well location

- Set the Inline-Xline to the location of the well track at the center of the analysis window. Position is set at the XY location where the trajectory crosses time at the center of the extraction window.



Selected wavelet

- Set the Inline-Xline to the location of the currently selected wavelet.

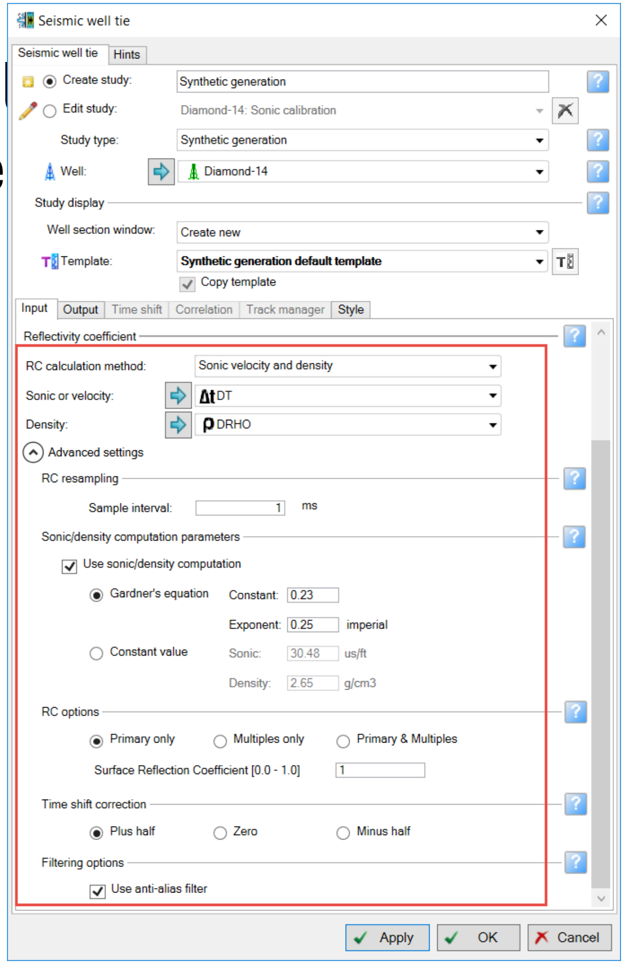


Active after the wavelet is extracted.

Position is set at IL/XL location where the wavelet was selected on the Predictability map.

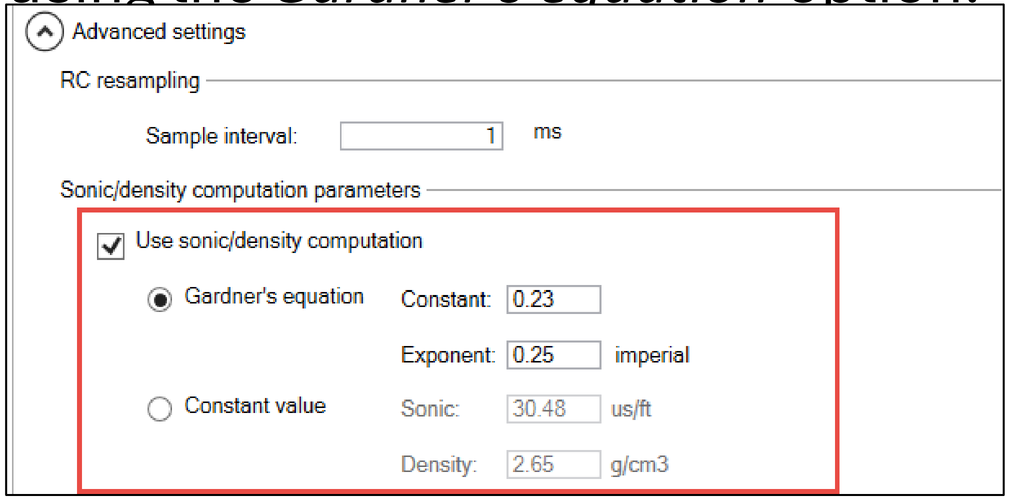
Reflectivity coefficient calculation

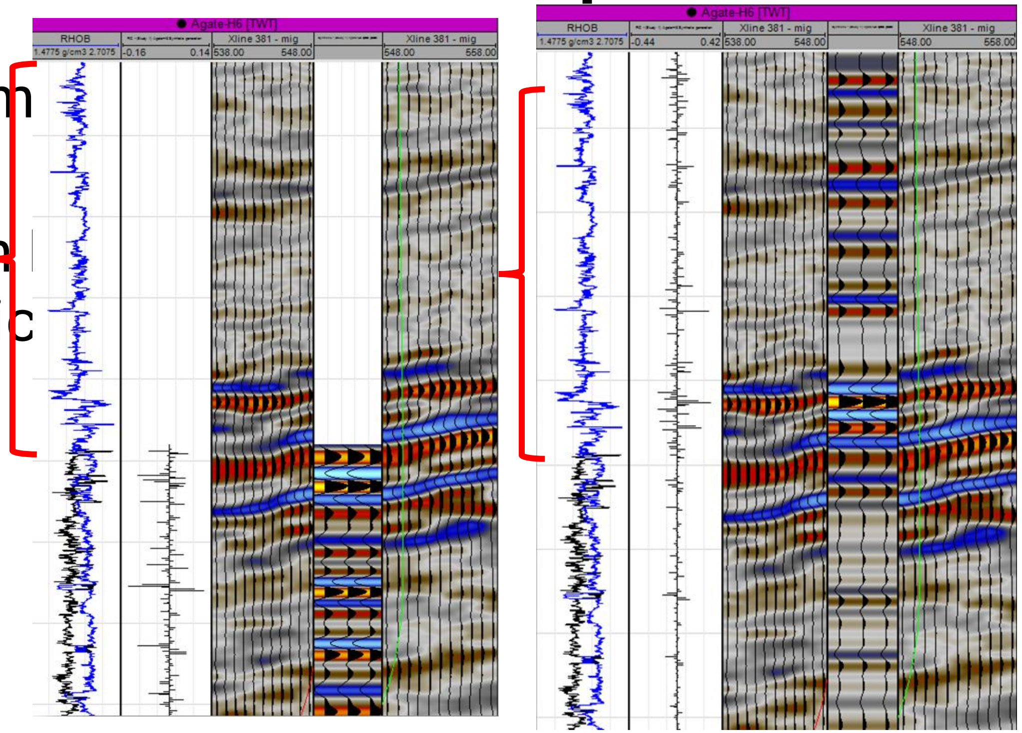
- Depending on the availability of data, the Refl coefficient can be calculated using these me
  - Impedance (AI/ EI/ EEI)
  - Sonic and density logs
  - Shear
  - Porosity
  - Aki and Richards PP
  - Aki and Richards PS
  - Any log



Gardner’s patching:
Auto-complete reflection coefficient inputs

- On-the-fly use of Gardner’s equation to com
DT or Density log during RC calculation.
- The patch shows that RC and Synthetic can
calculated for a missing section of the Sonic
using the *Gardner’s equation* option.

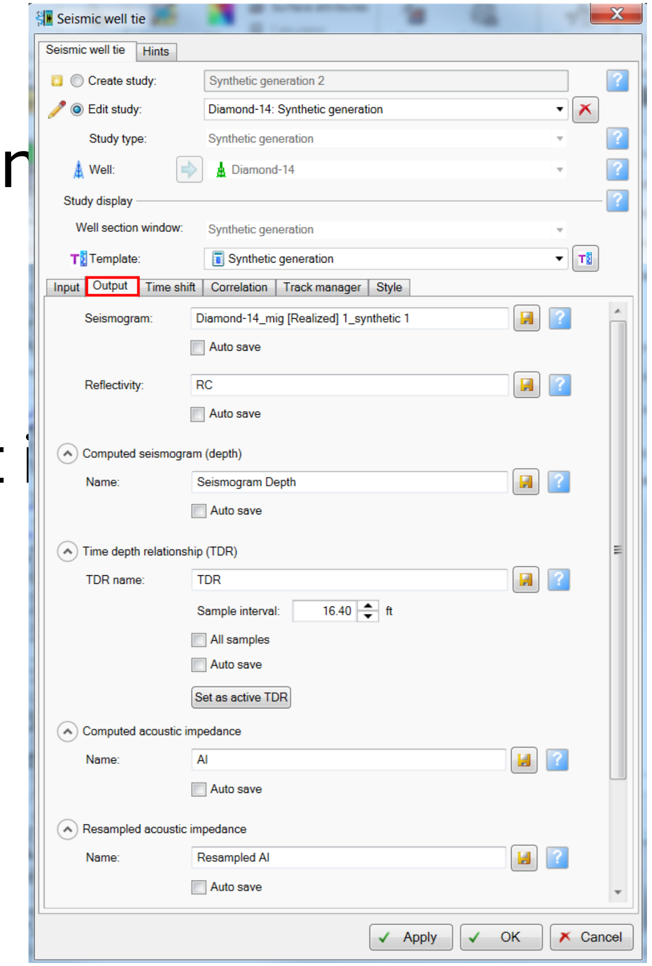




Output tab

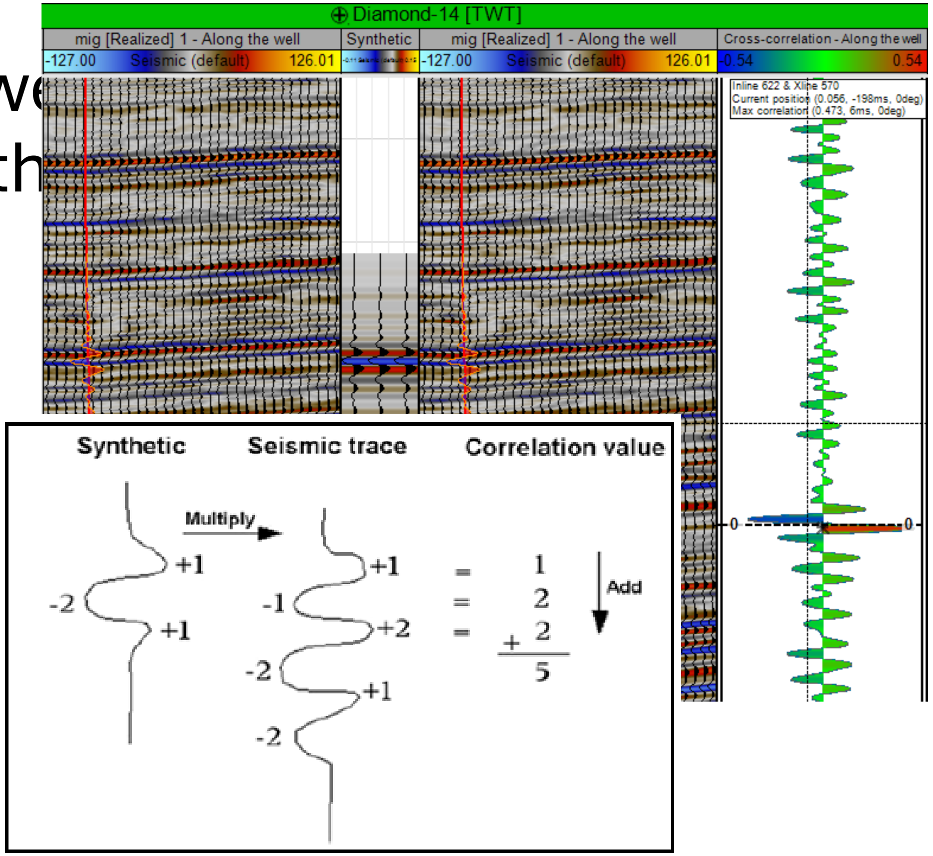
- Use the **Output** tab in the Synthetic generation control the output results from the workflow.

- To change the name of any output and save it i **Input** pane, click *Save* .



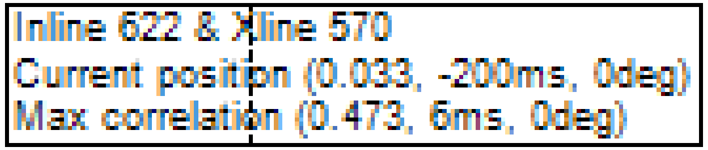
Correlation track (1)

■Calculates the cross-correlation betwe
synthetic and seismic by calculating th
shift to be applied to the synthetic.



■This correlation value is the sum of th
products of the synthetic and seismic
amplitudes at each position in time.

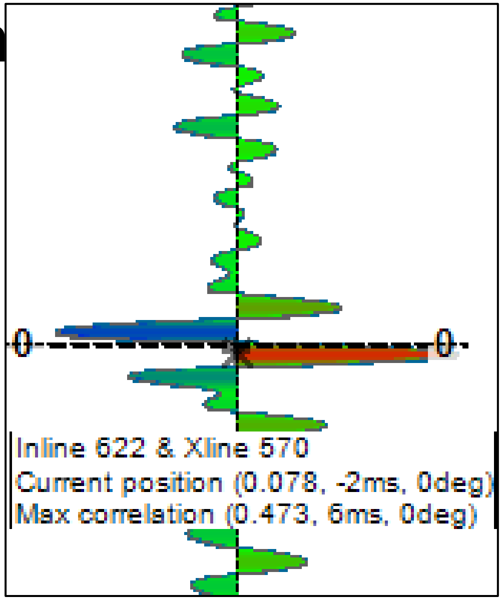
Correlation track (2)

- When you place the cursor over the Correlation track, you can read the inline and crossline positions, the correlation value that corresponds with the mouse position (Current position), and the Max correlation from the box labeled in the upper right side of the track 

- The lag value at any point in the Correlation track is the time shift applied to the synthetic to move it into the position where it was when the correlation values were calculated.

Correlation track (3)

A dotted line marks the trace posted on the Correlation track with the point of highest correlation. The lag value at this point is the amount of bulk shift that you would need to apply to move the synthetic into this best ma

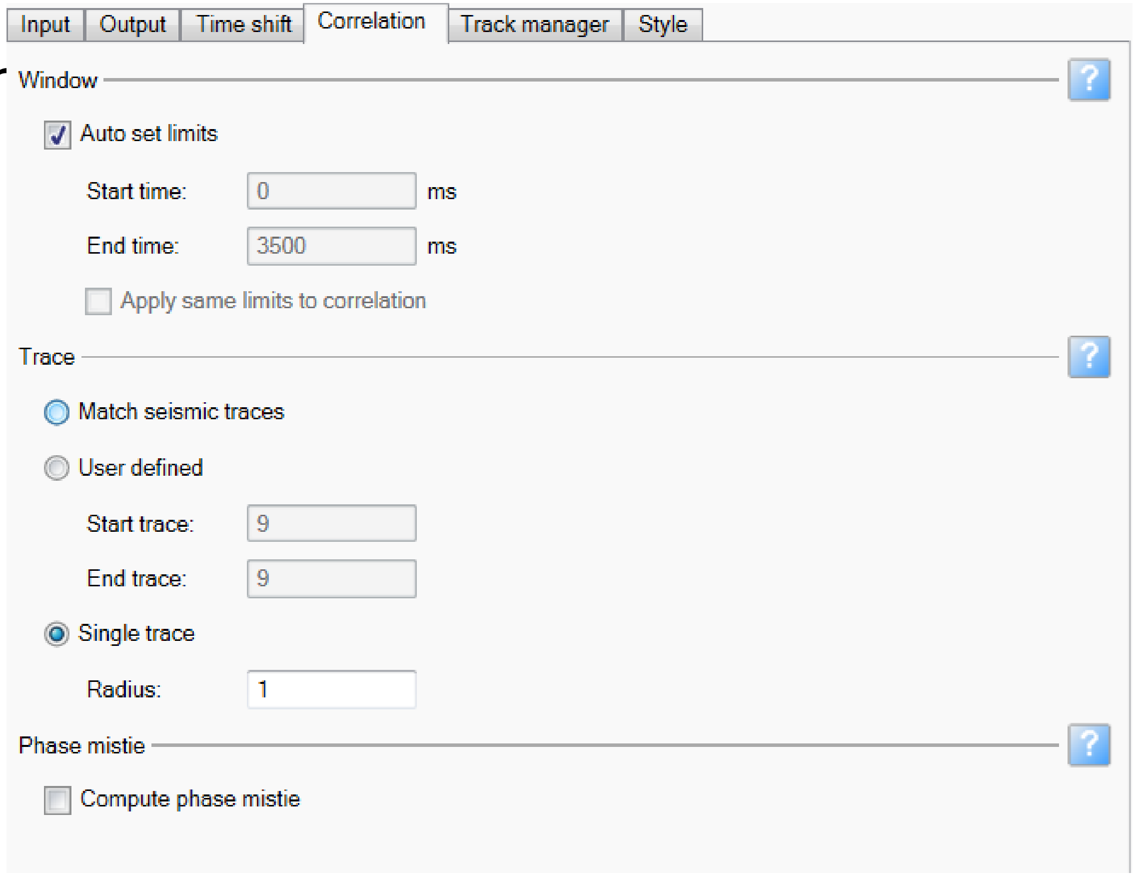


Correlation tab (1)

This tab controls the display param
of the Correlation track.

It has three main sections:

- Window
- Trace
- Phase mistie



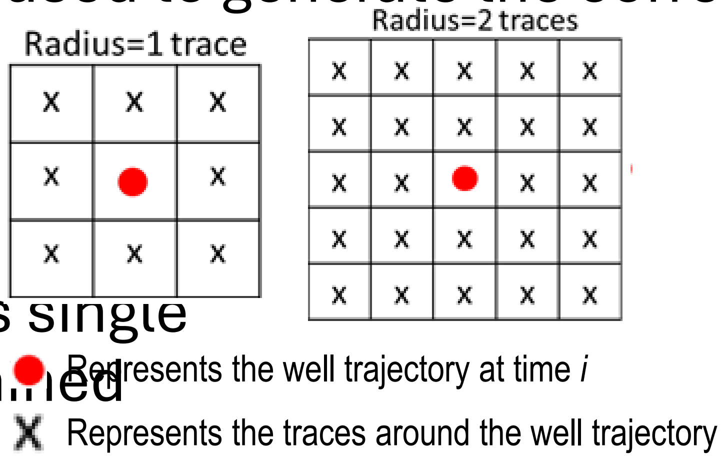
Correlation tab (2)

- **Match seismic trace:** Visualize the generated correlation traces using the entire range of the reference seismic, defined in the **Seismic display** section of the **Seismic well tie** dialog box.

User defined: Specify the range of traces to be used to generate the correlation.

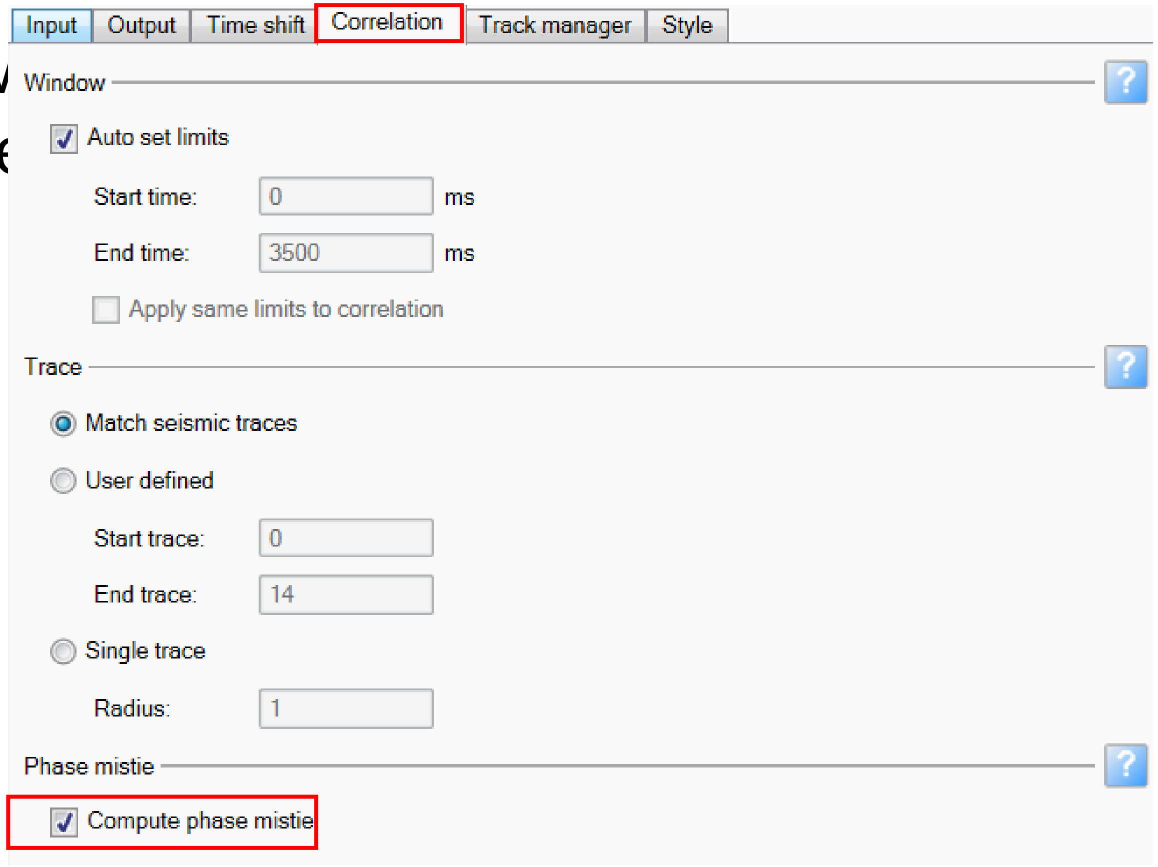
Single trace:

- A single trace is calculated following the well trajectory.
- The traces used in the calculation for this single trace (average trace) are the traces contained within a radius that you determine.



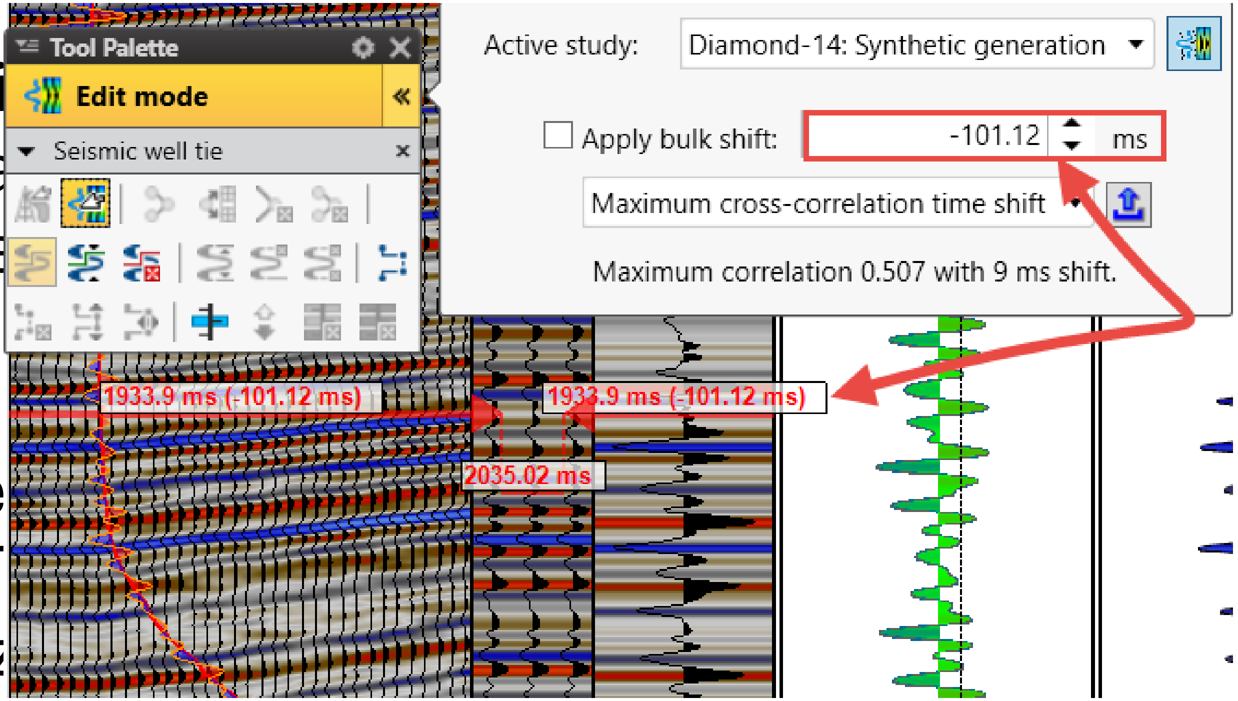
Correlation tab (3) Phase mistie

- Calculates the correlation betw synthetic/seismic while conside phase of the wavelet.



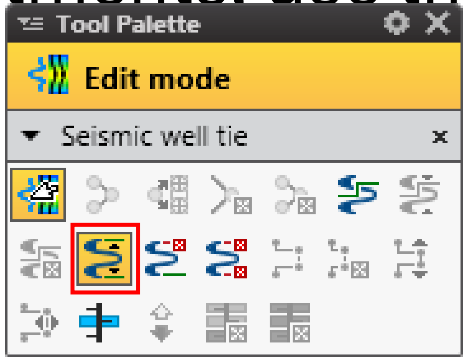
Use interactive bulk shift

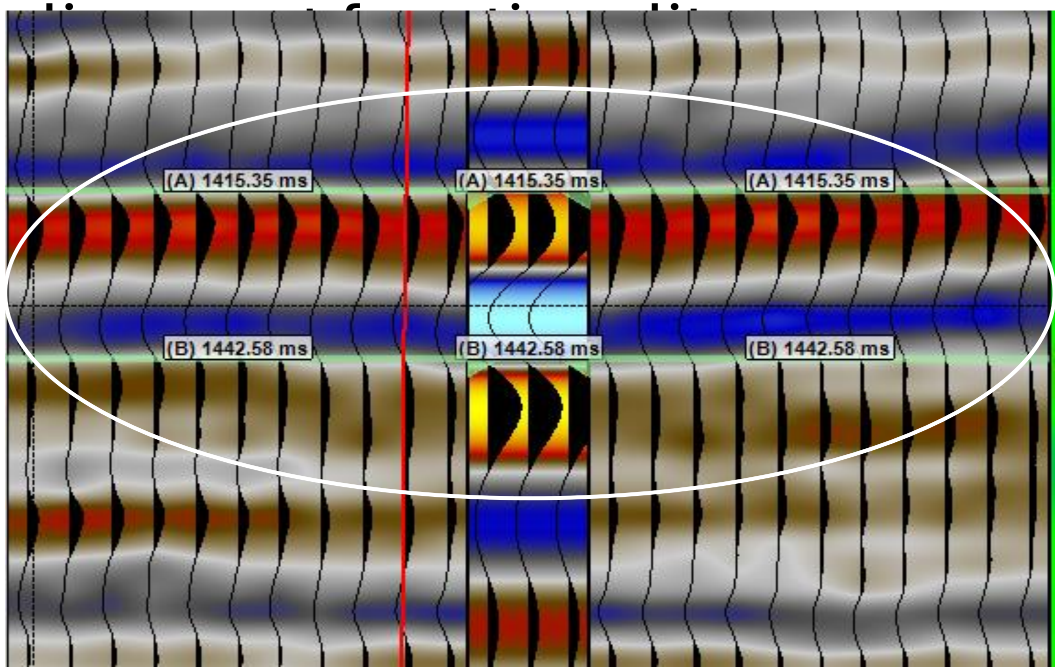
- The next step is to do a bulk shi
the synthetic to get a closer ma
between the synthetics and the
seismic.
-
- If the quality of the tie is good e
a stretch and squeeze will NOT
necessary. (Stretch and squeez
not recommended as the first
approach.)



Use continuous alignment

- Sometimes, small adjustments must be made to match the synthetics with the seismic after the bulk shift is implemented. To perform these adjustments, use the
the process is intera

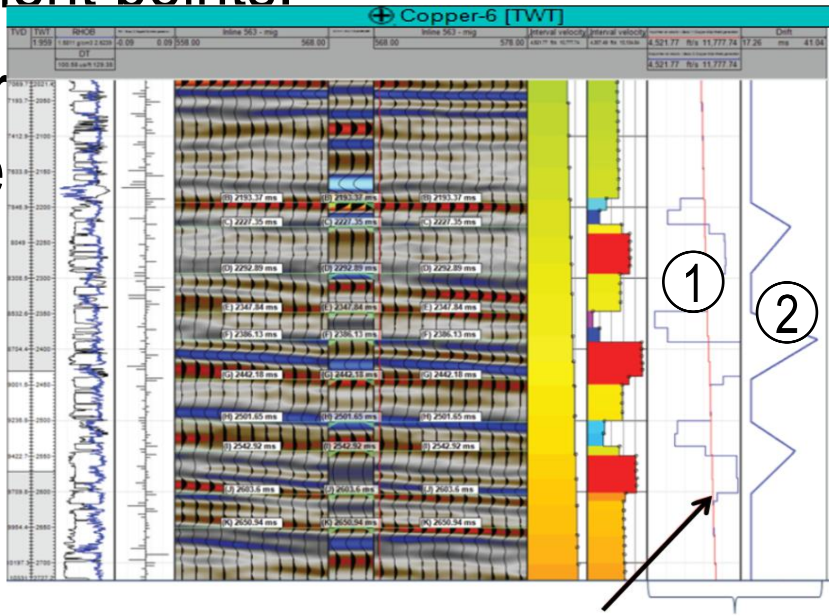




Note: You can add as many alignment points as required.

QC interval velocity

- The Input Interval Velocity vs. Output Interval Velocity and the Drift curve in time tracks show information about the resultant interval velocity and the time shift applied to the alignment points.
- These tracks act as guides for the synthetic generation proce and velocity editing.

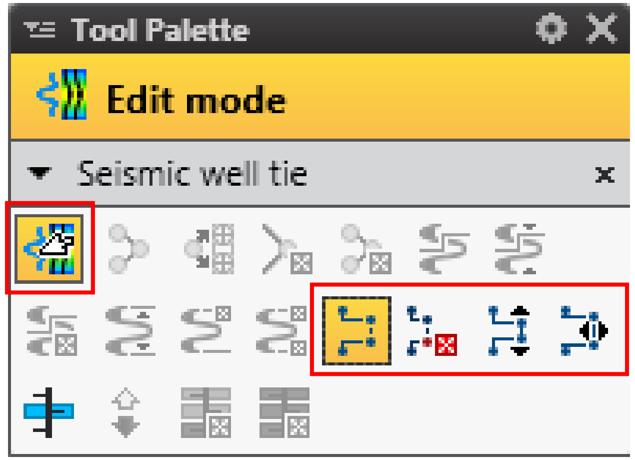


1 Input Interval Velocity vs. Output Interval

2 Drift Curve in time for shift applied

Interval velocity manipulation (1)

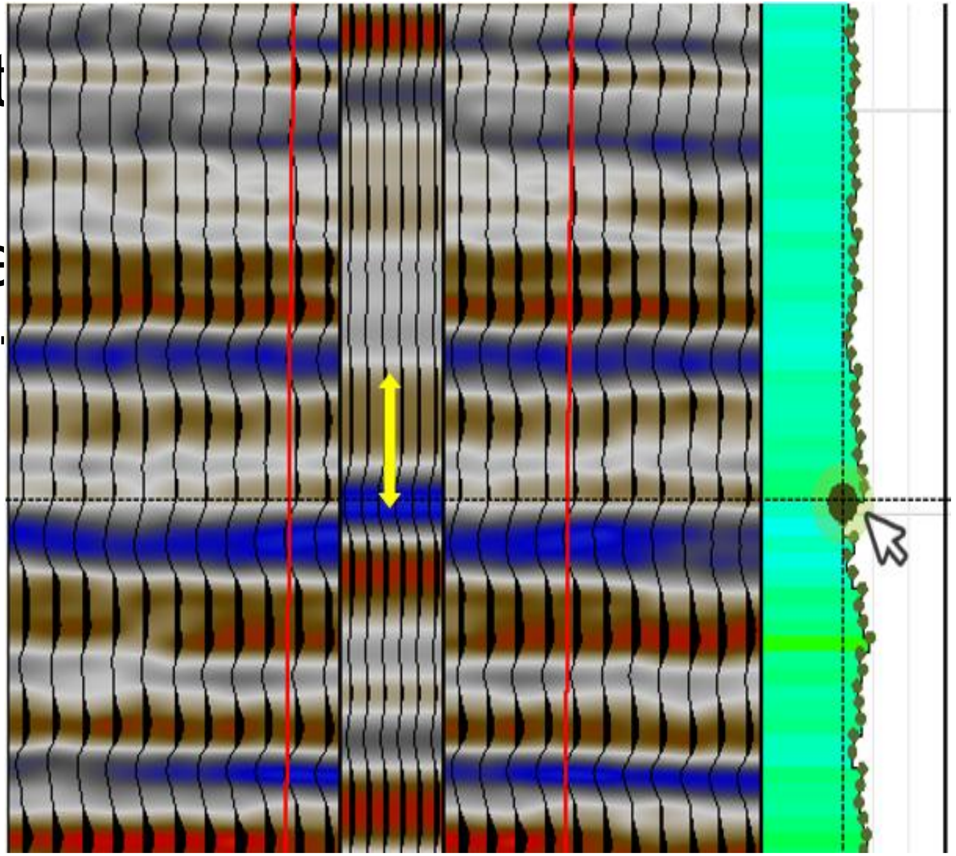
- Edit/manipulate the interval velocity on the fly to check the impact on the data.
- Edit the velocity values visually in a **Well section window**.
- Use the **Seismic well tie Tool Palette** to manipulate and delete velocity points.
- Lock the velocity manipulation into vertical and horizontal directions.



- Manipulate interval velocity
- Delete interval velocity knee
- Vertical interval velocity manipulation
- Horizontal interval velocity manipulation
- Edit mode

Interval velocity manipulation (2)

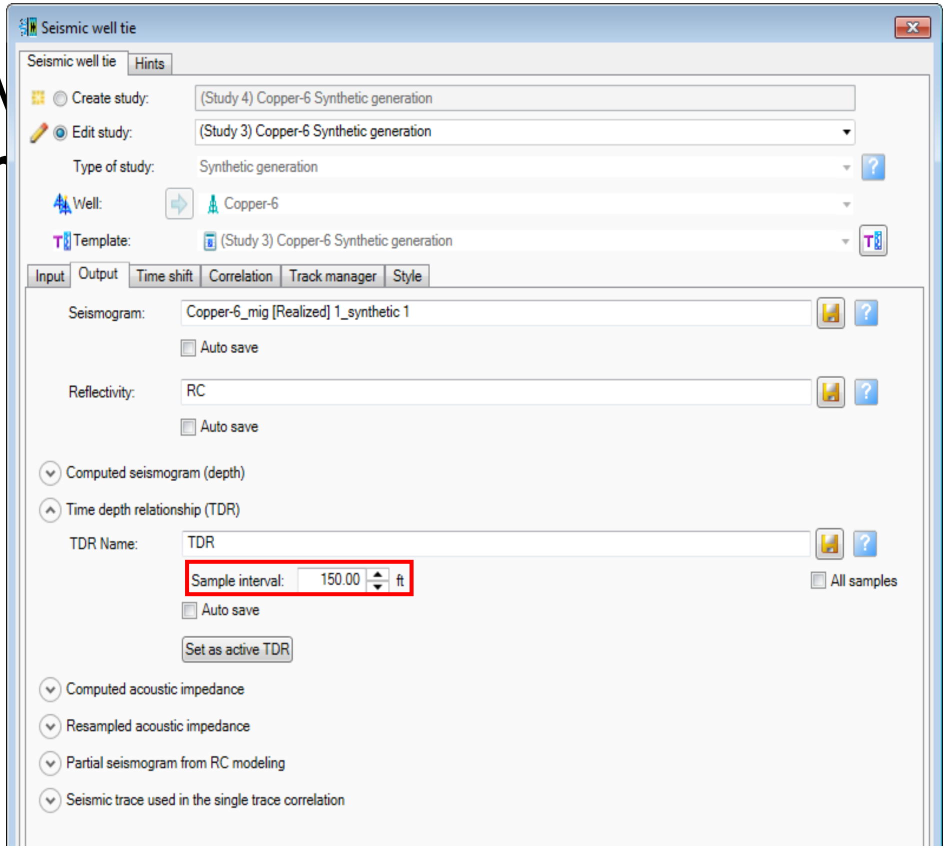
- In the Interval velocity track, you can manipulat  the points when the Seismic well tie *Edit mode*  and *Manipulate interval velocity* buttons are switched on in the **Seismic well tie Tool Palet**



Interval velocity manipulation (3)

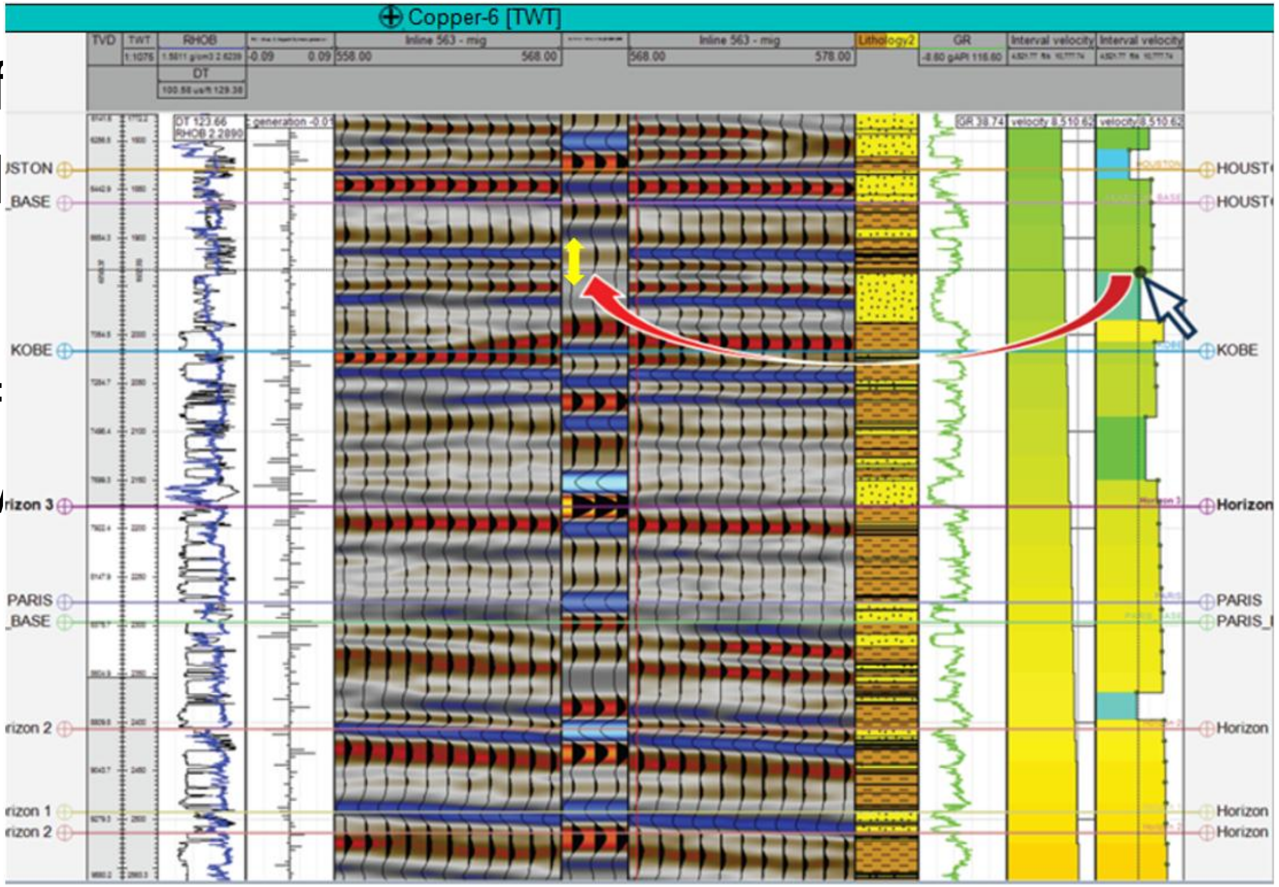
- You can set the sampling of the interv velocity on the **Output** tab of the **Seisn well tie** dialog box in the **Time/depth relationship (TDR)** section.

- Remember that the interval velocity i attribute of the checkshot.



Interval velocity manipulation (4)

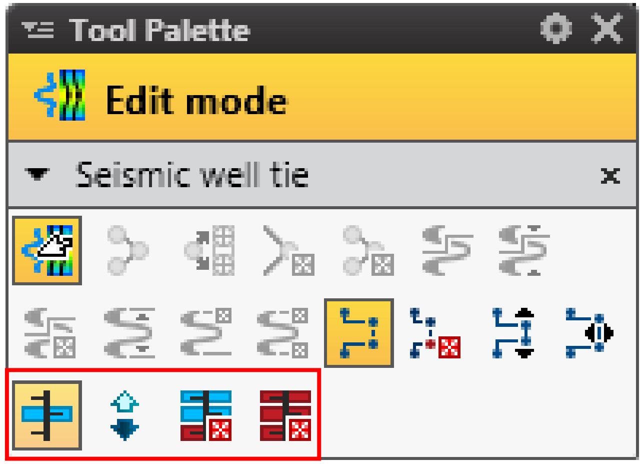
- The interactive manipulation f
  can be an important tool for q
  control purposes, especially i
  have information about the lit
  (from the driller or geologist, f
  example) because you know v
  range of velocity to expect.

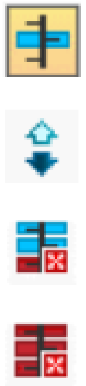


Reflection coefficient (RC) modeling (1)

The RC modeling process allows you to:

- select/deselect reflection coefficients and see the impact in the new synthetic created through RC modeling
- study tuning effects
- filter problematic areas from your synthetic.





Activate RC modeling

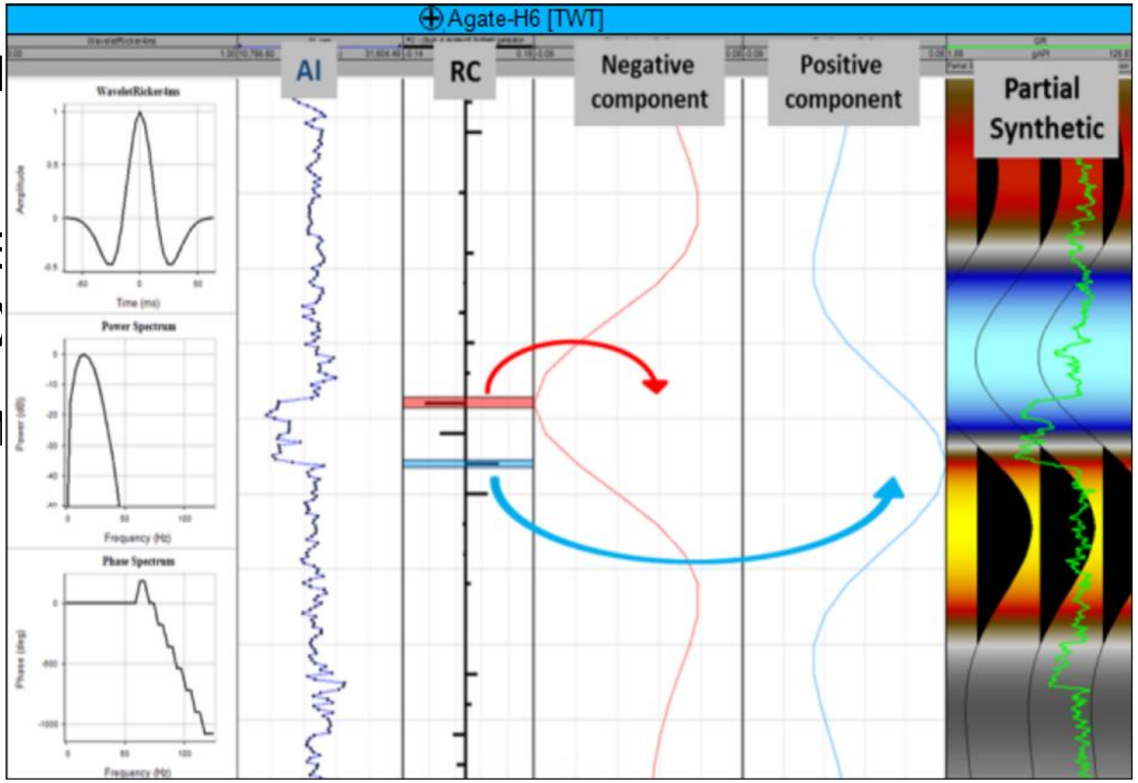
Invert the RC selection

Delete single RC selection

Delete all RC selections

Reflection coefficient (RC) modeling (2)

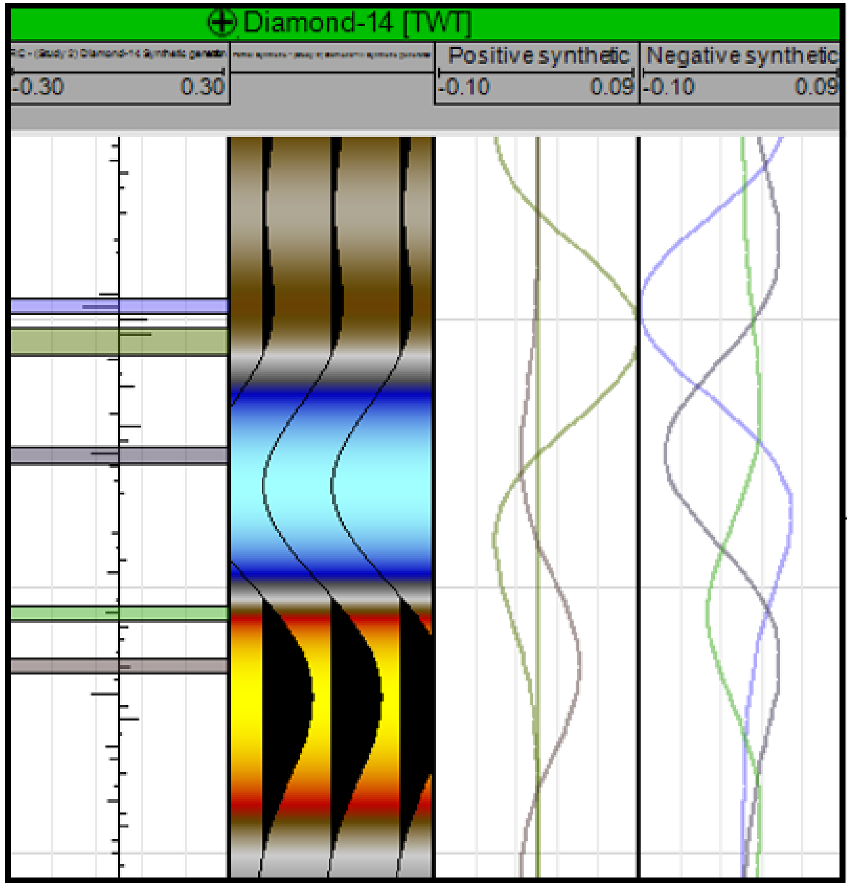
- The result appears in three tracks i
  Well section window:
  - ■Partial Synthetic (represents the synthetic from your selected RC
  - ■Positive Synthetic (positive com of your selected RC values).
  - ■Negative Synthetic (negative component of your selected RC



Run Reflection coefficient (RC) modeling (1)

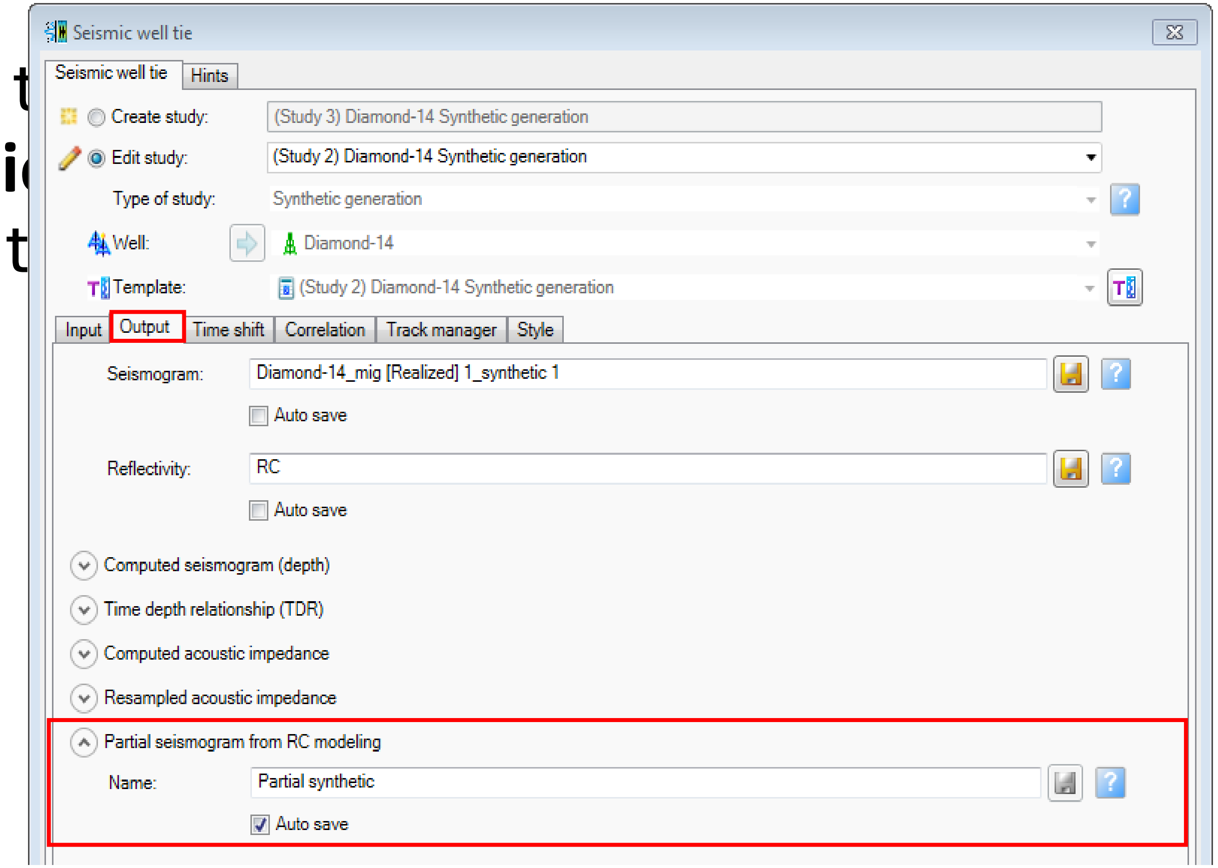
1. In the **Seismic well tie Tool Palette**, click *Activate RC modeling* .
   Three more tracks appear in the Synthetic of the **Well section window** (WSW):
   – Partial Synthetic
   – Positive component
   – Negative component

2. From the RC series track, select/clear area your log to see the impact on your synthet



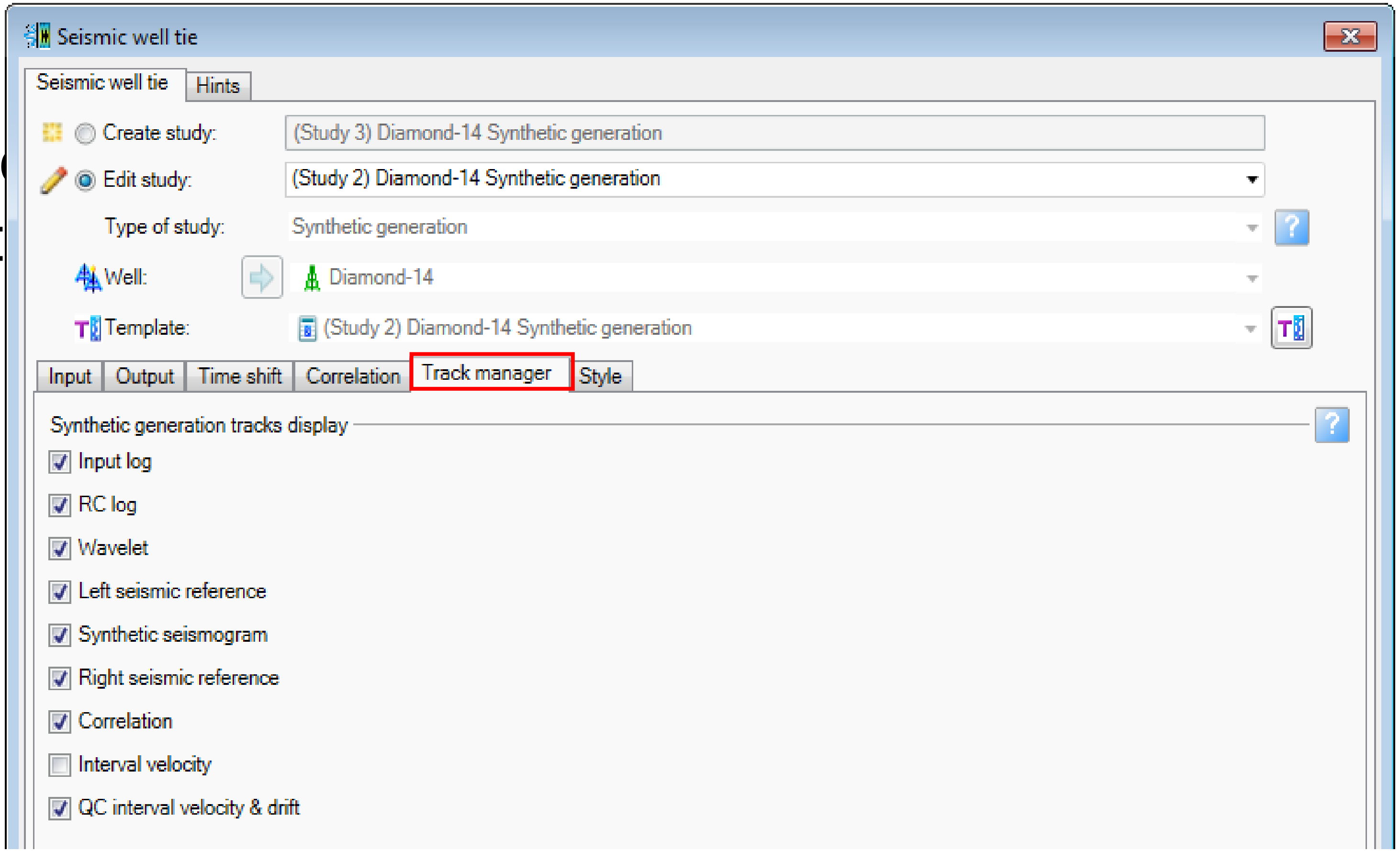
Run Reflection coefficient (RC) modeling (2)

3. Save your partial synthetic. On t **Output** tab of the **Seismic well ti** dialog box, click *Save* or select t *Auto save* check box.



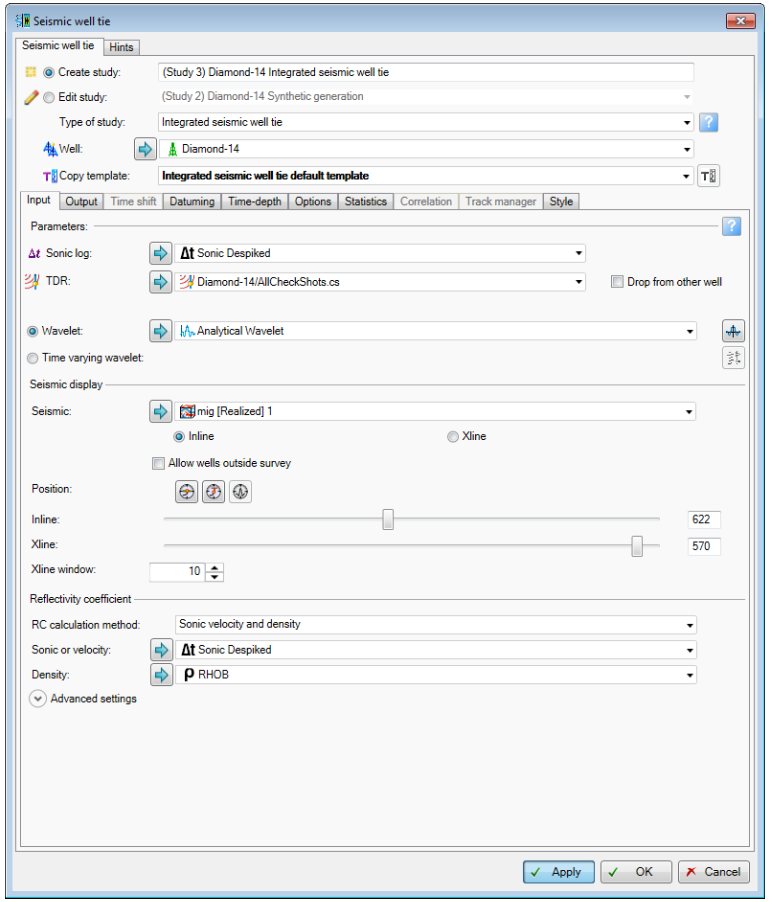
Track manager

- The **Track manager** tab provides a shortcut to h you manage the tracks t you want to show in the study.



Integrated seismic well tie

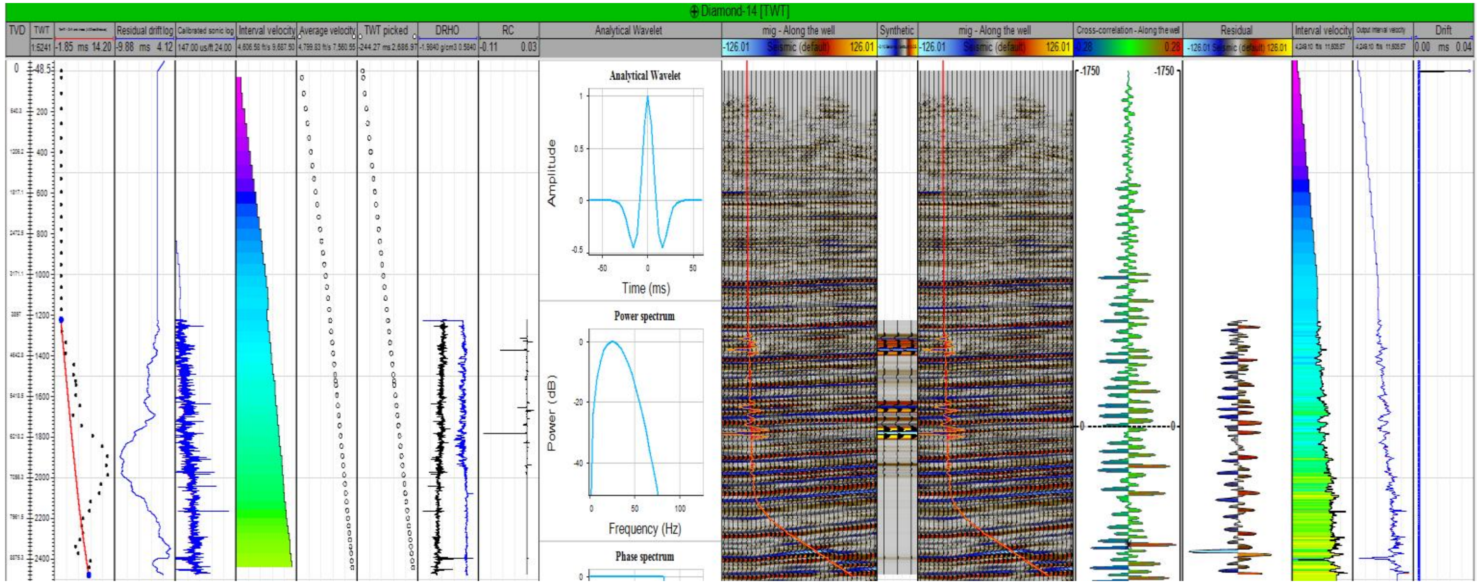
- An integrated process in which Sonic Calibrat
Synthetic Generation is done simultaneously
same **Well section window**.



Integrated seismic well tie study template

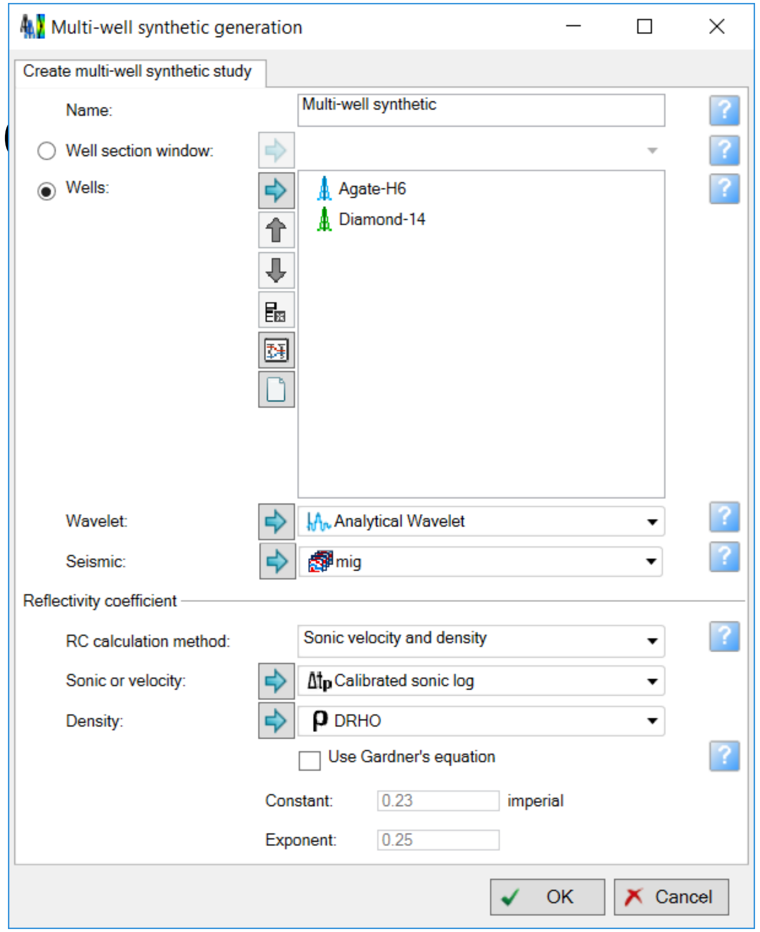
- Similar to the Sonic calibration and Synthetic generation, you can select a template for the study or use the default template.

Sonic calibration	+ Synthetic generation



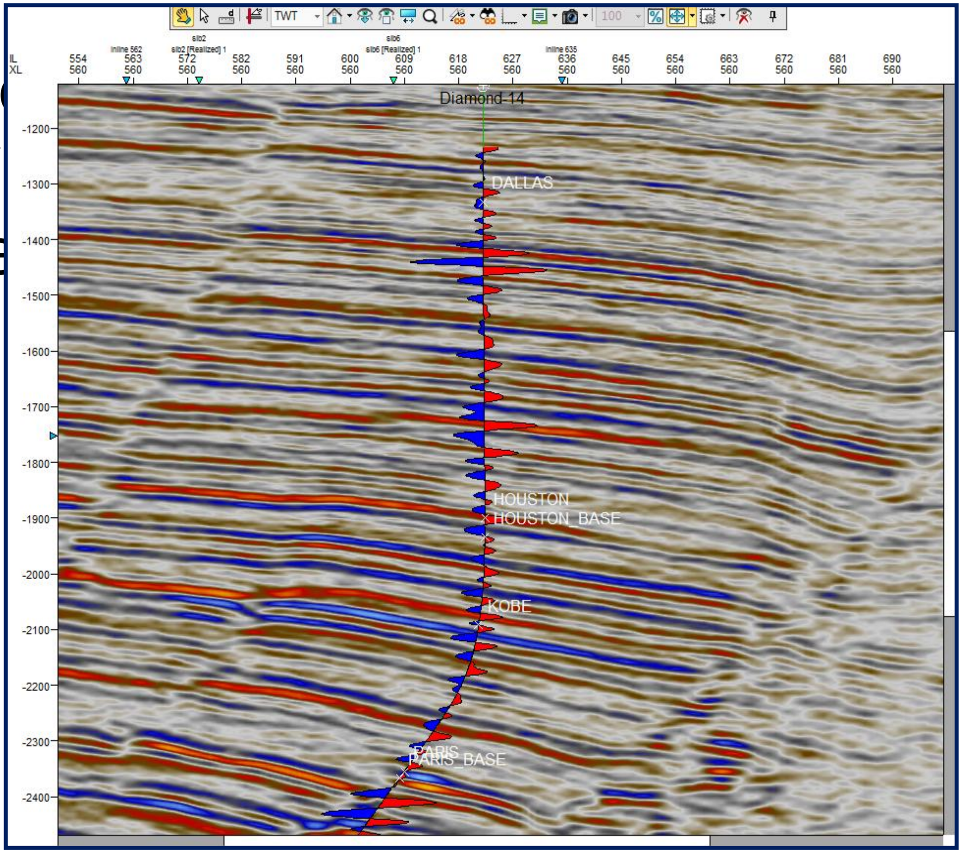
Multi-well synthetic seismogram

- You have the option to create a synthetic generation study for multiple wells in a single **Well section window**.



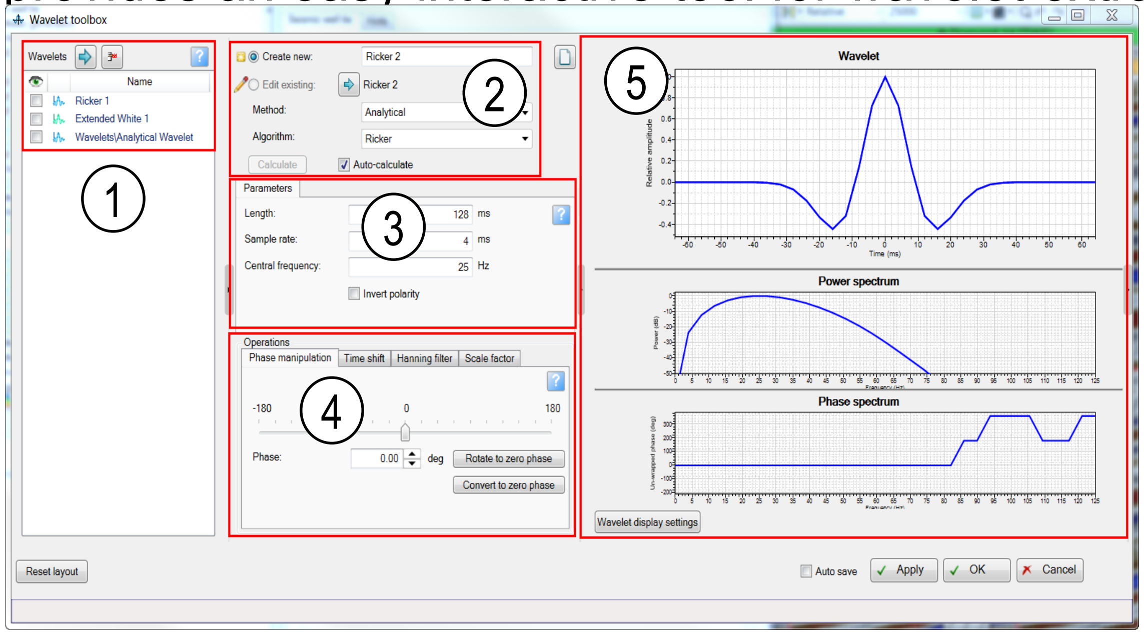
Synthetic seismogram in the Interpretation window

- In an **Interpretation window**, you can sh
  the wellbore trajectory with the syntheti
  seismogram to identify the seismic refle
  that represent the geological well tops (
  before starting horizon interpretation.



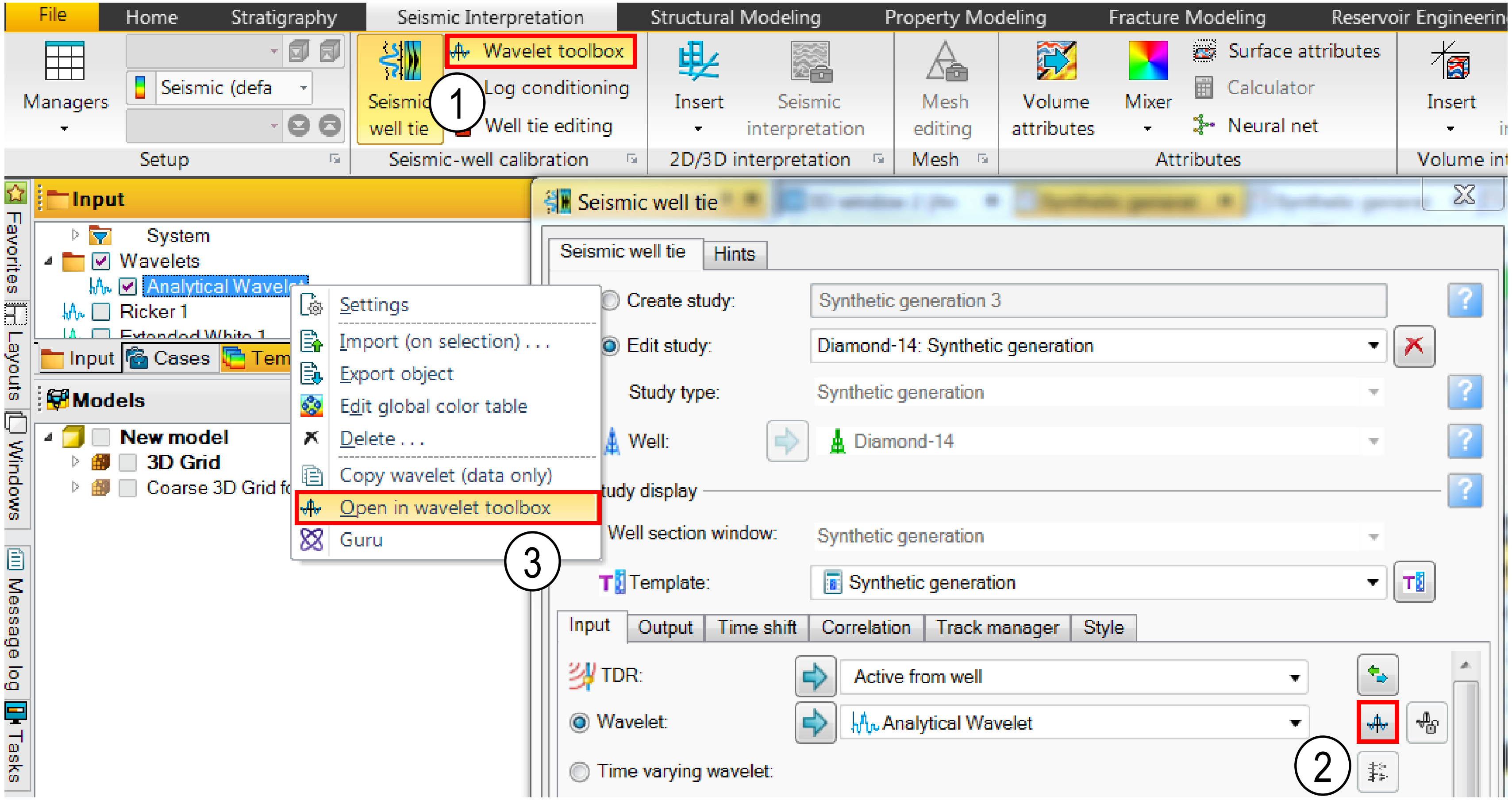
Wavelet toolbox

- The **Wavelet toolbox** integrates all related processes (Wavelet extraction, Wavelet Builder, and Wavelet viewer) in a single canvas.
- It provides an easy interactive tool for wavelet extraction.



1 Displayed Wavelet list
2 Method and algorithm
3 Parameters of extraction/generation
4 Operations
5 Visualizations

Access the Wavelet toolbox



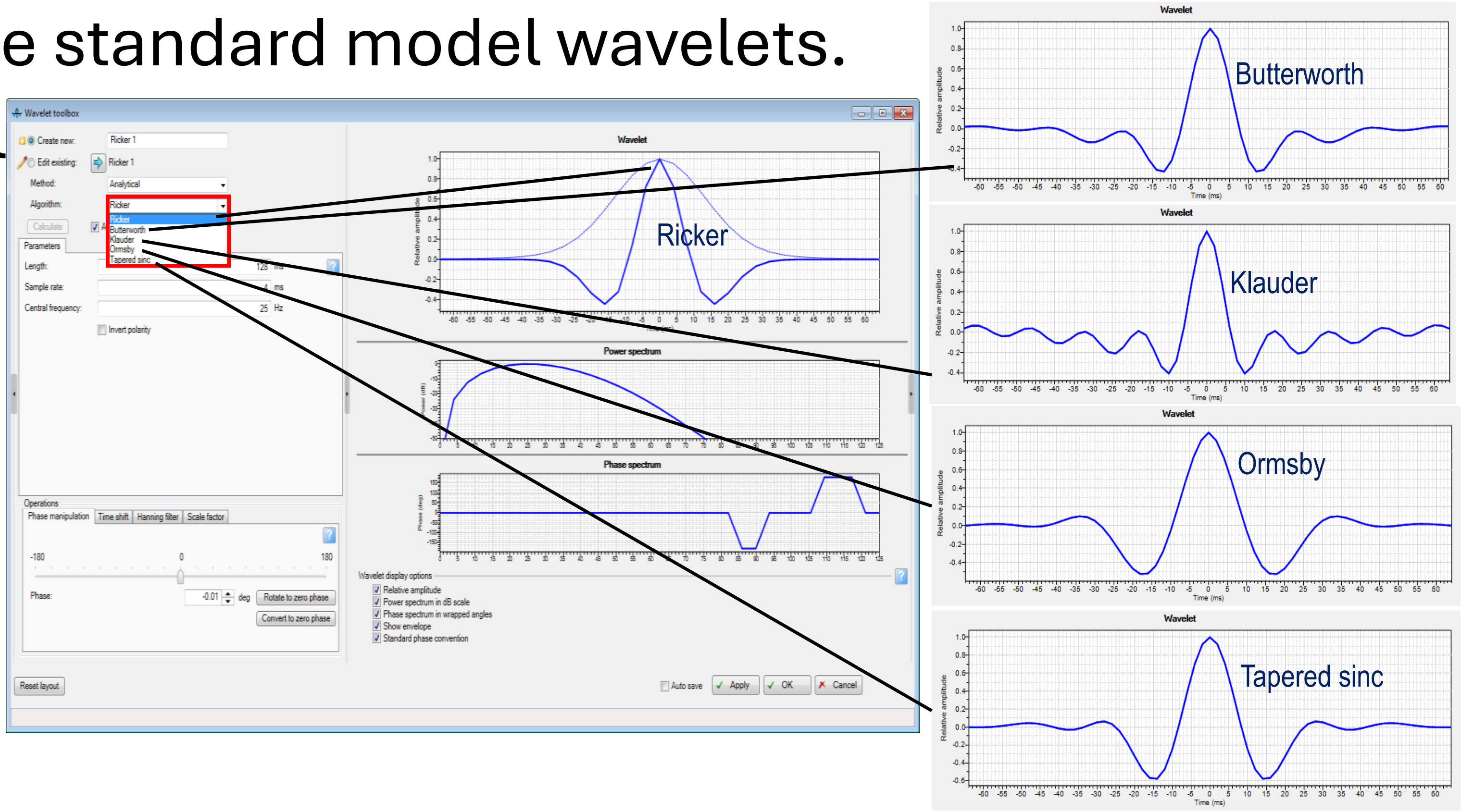
Types of wavelet extraction methods

- Analytical:
  - Ricker
  - Butterworth
  - Klauder
  - Ormsby
  - Tapered Sinc

- Statistical:
  - Extraction
  - Extended extraction
- Deterministic:
  - Extended White
  - Signal/Noise
  - Isis Frequency
  - Isis Time
- Multi wavelet: Wavelet Average
- Multi well

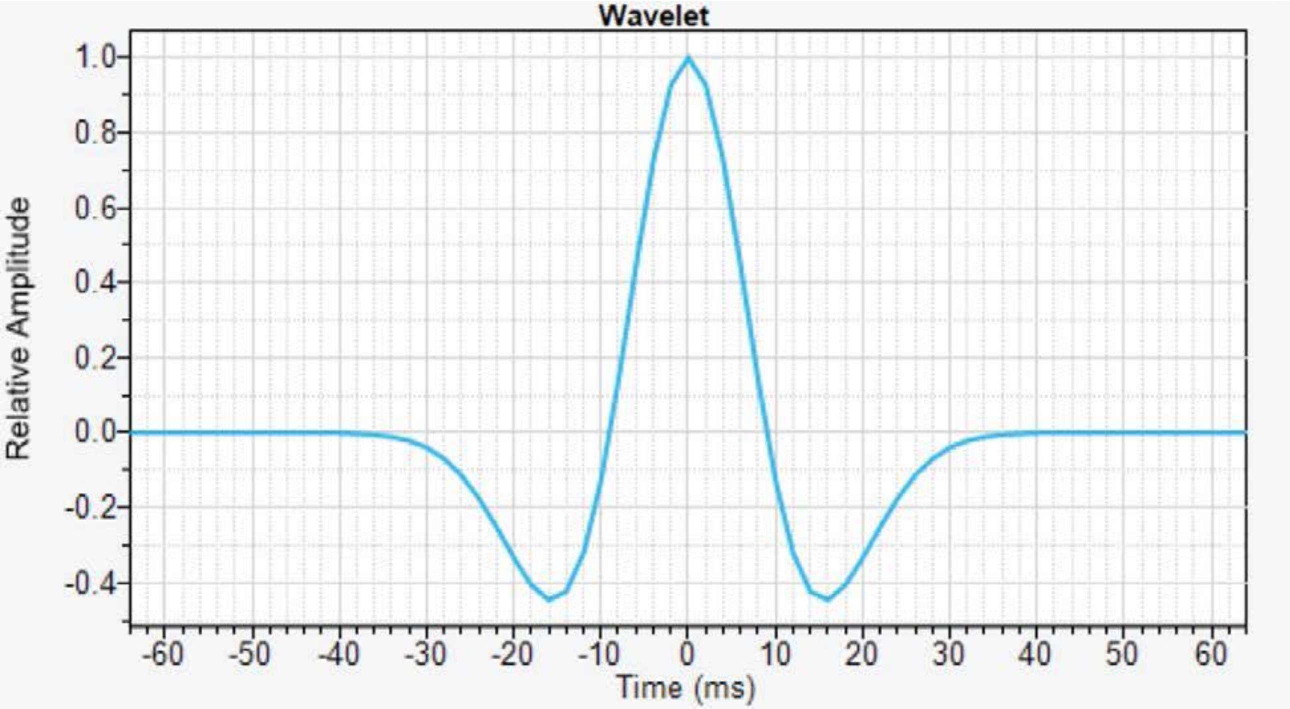
Analytical method

- Analytical wavelets are standard model wavelets.
- Five types are availabl



Ricker

- A Ricker filter requires only the peak frequency.
- This filter commonly is used for synthetic modeling.
- No bandpass filter is involved; t frequency and phase spectrums are purely a function of the pea frequency input.



Ormsby

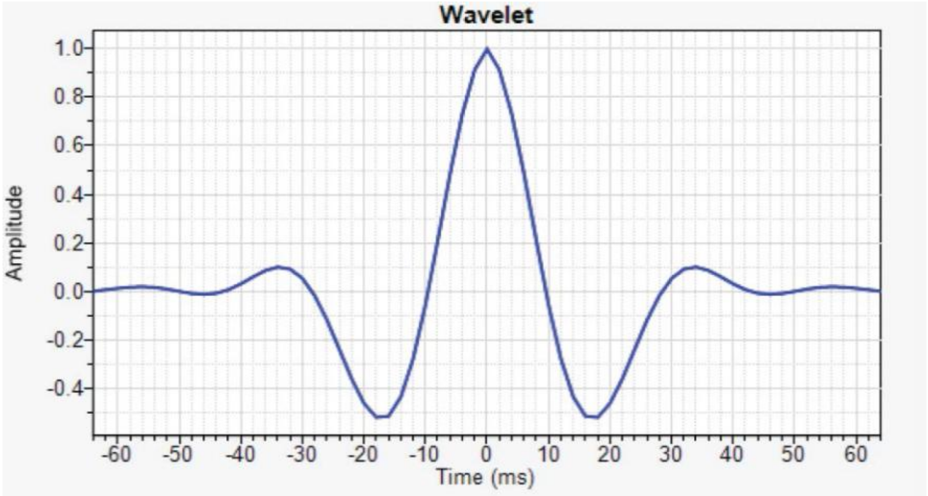
- The bandpass of an Ormsby filter can be described by using as many as four corner frequencies.

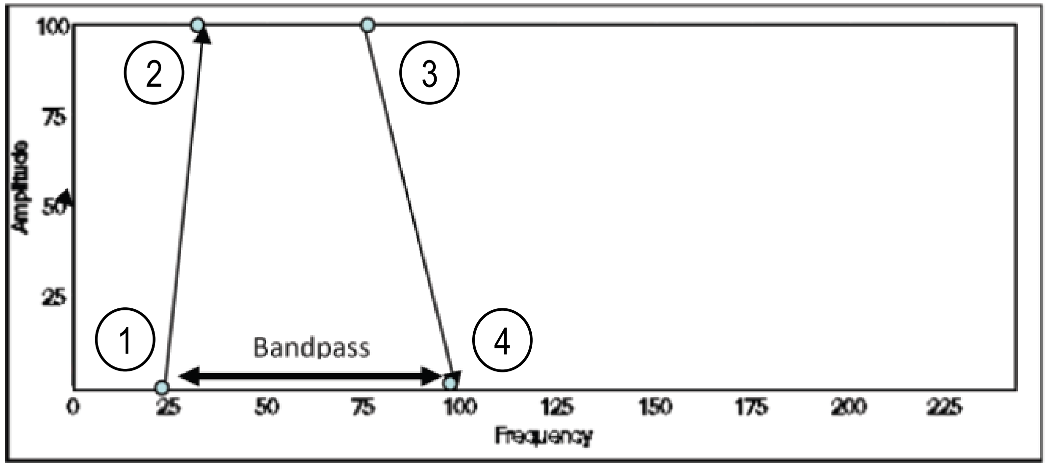
1 Low cut frequency

2 Low pass frequency

3 High pass frequency

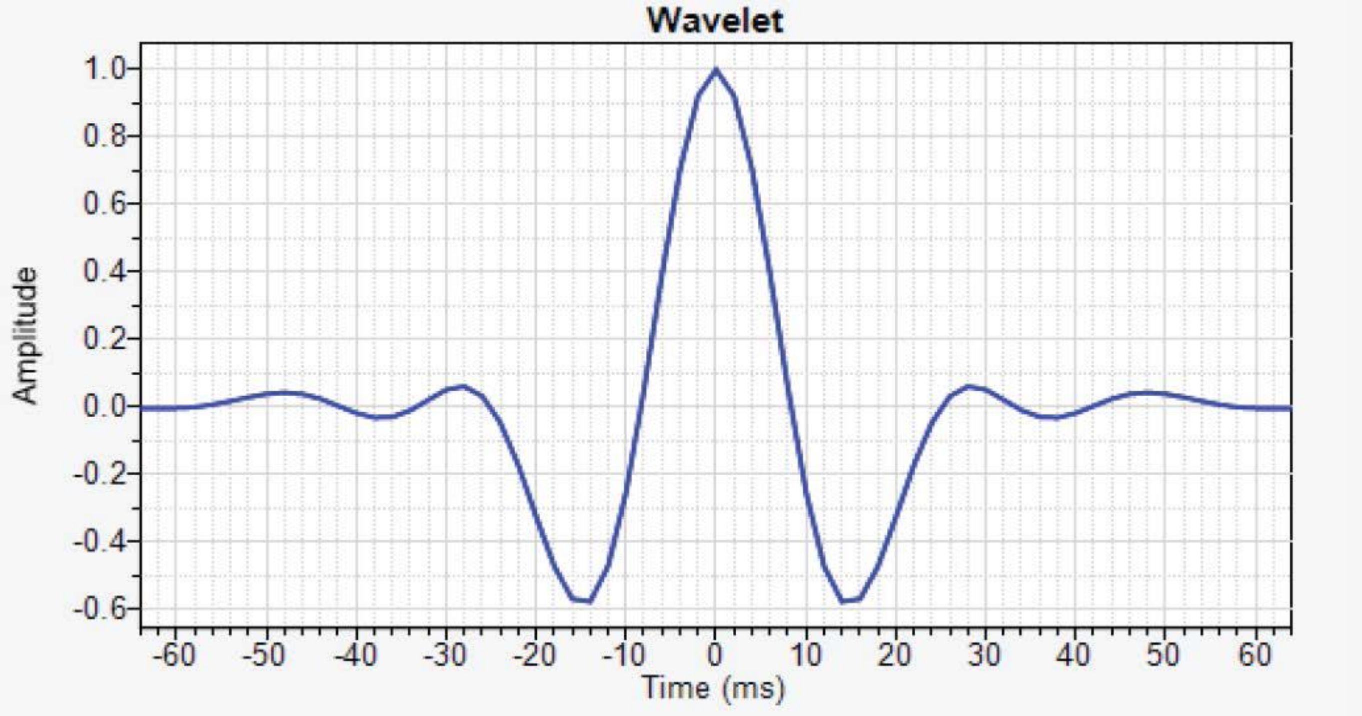
4 High cut frequency





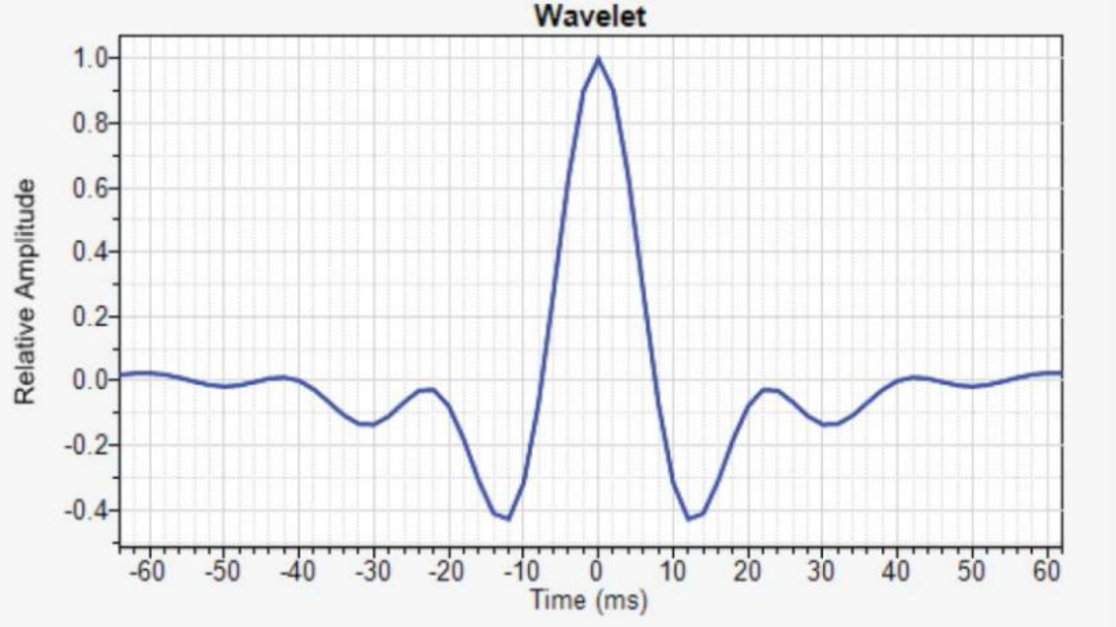
Tapered sinc

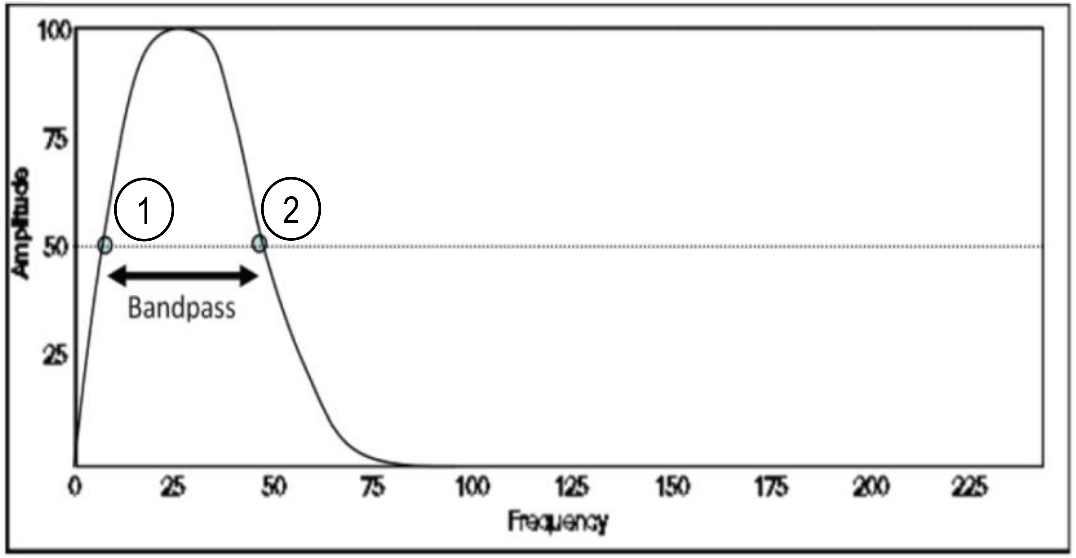
- A tapered sinc filter is defined by a low and a high cutoff similar to a Butterworth filter, but it applies no further filters. A Butterworth filter applies two slopes.



Butterworth

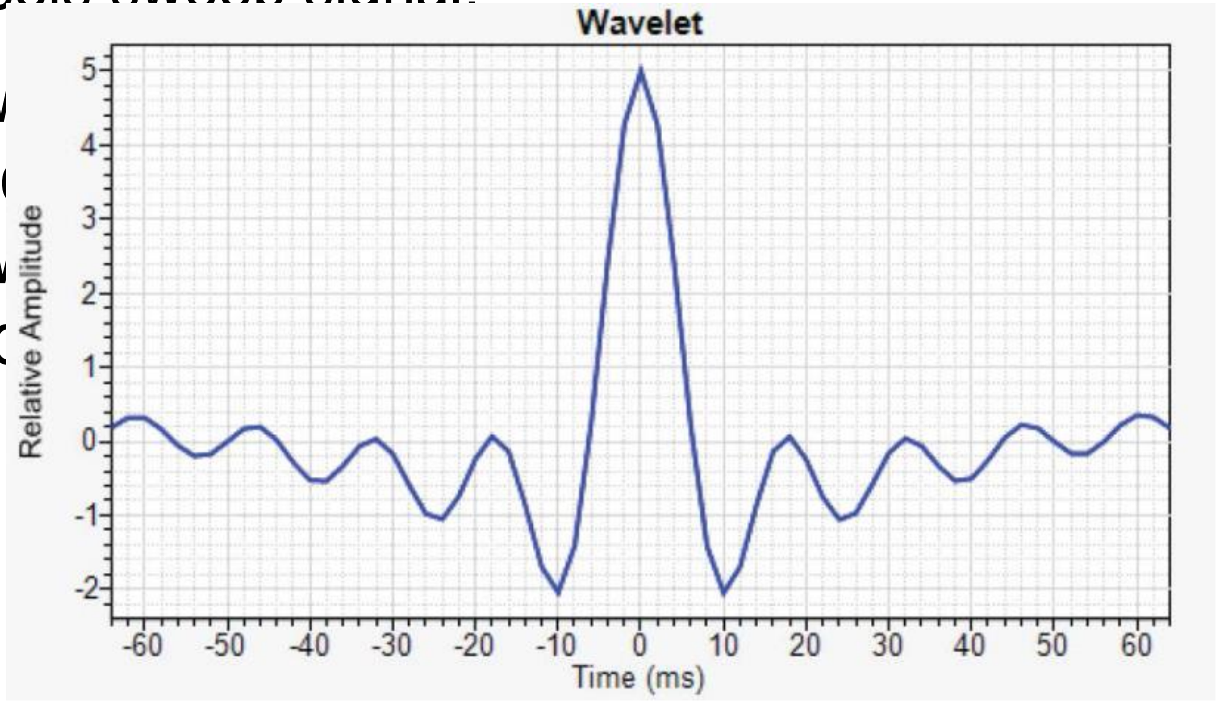
- The Butterworth bandpass consists of two cutoff frequencies taken at 3dB down from maximum power, or approximately half power (~50% on the amplitude scale). In the figure, frequencies are at 10Hz and 50Hz. The Butterworth filter also requires two slopes.





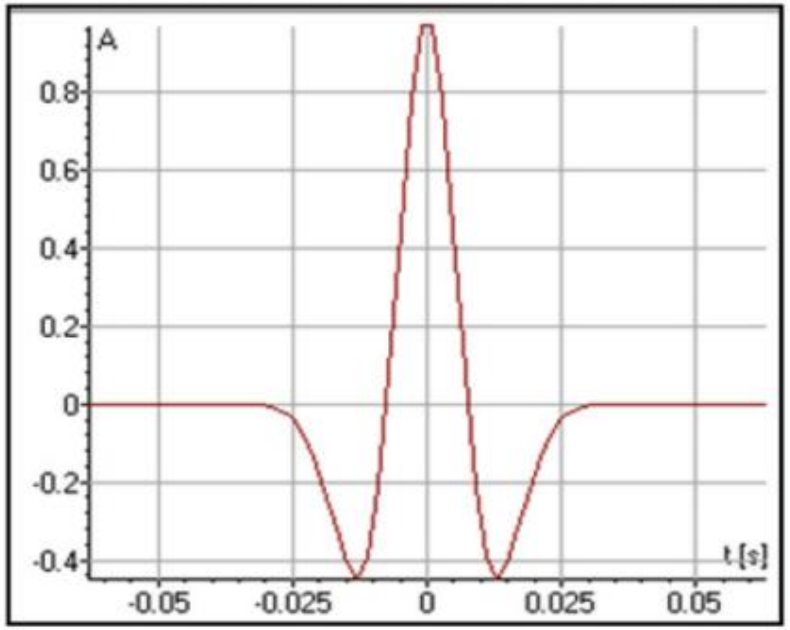
Klauder

- An analytical approximation of the Klauder wavelet is computed through autocorrelation of an actual vibroseis sweep signal.
- A Klauder wavelet is defined by tw
frequency cutoff values: a low cut
and a high cutoff. The figure show
these frequencies set at 10Hz anc
70Hz.

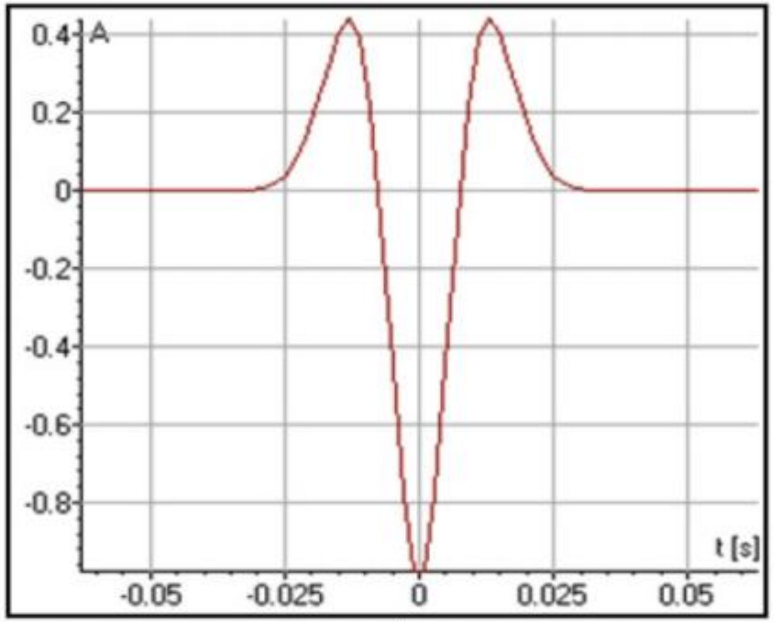


Wavelet phase

- Convention for a zero-phase wavelet for the USA and Europe are opposite in phase



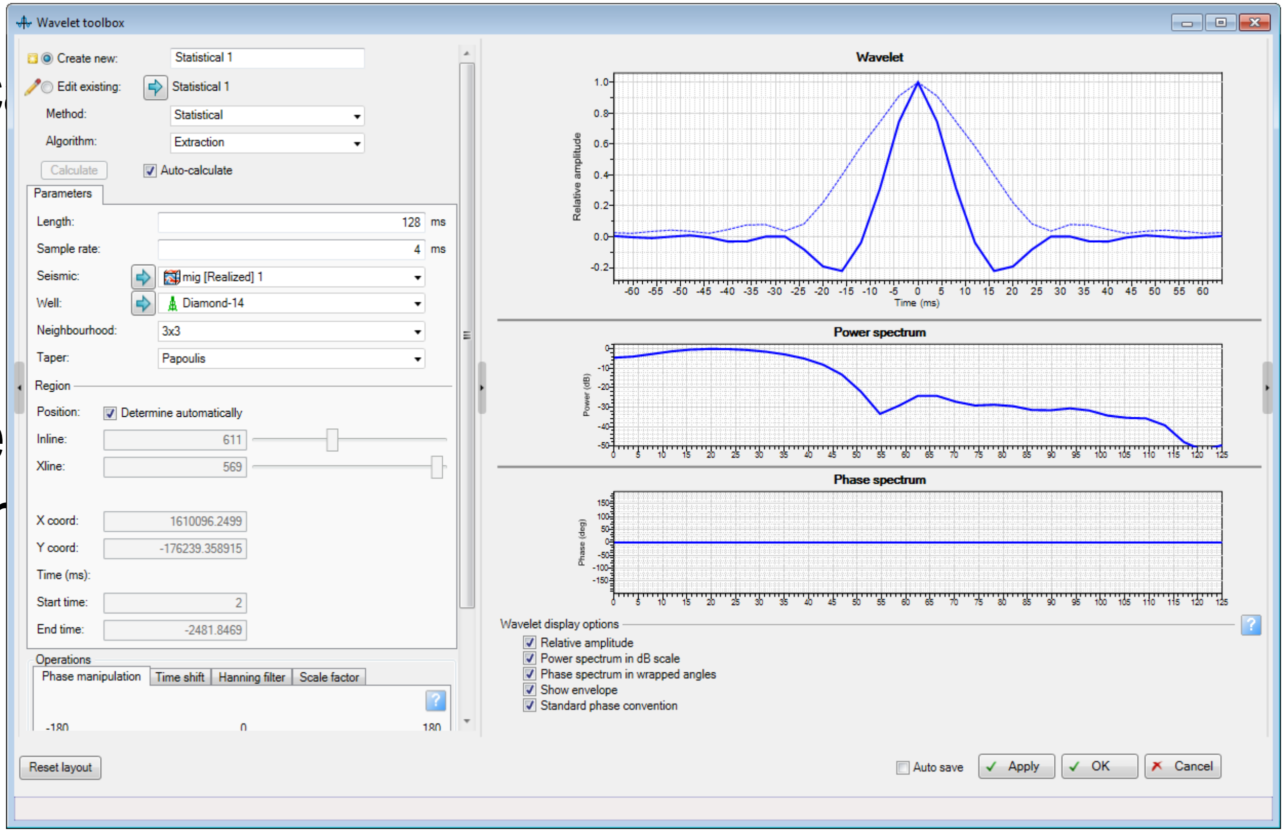
USA zero-phase wavelet



European zero-phase wavelet

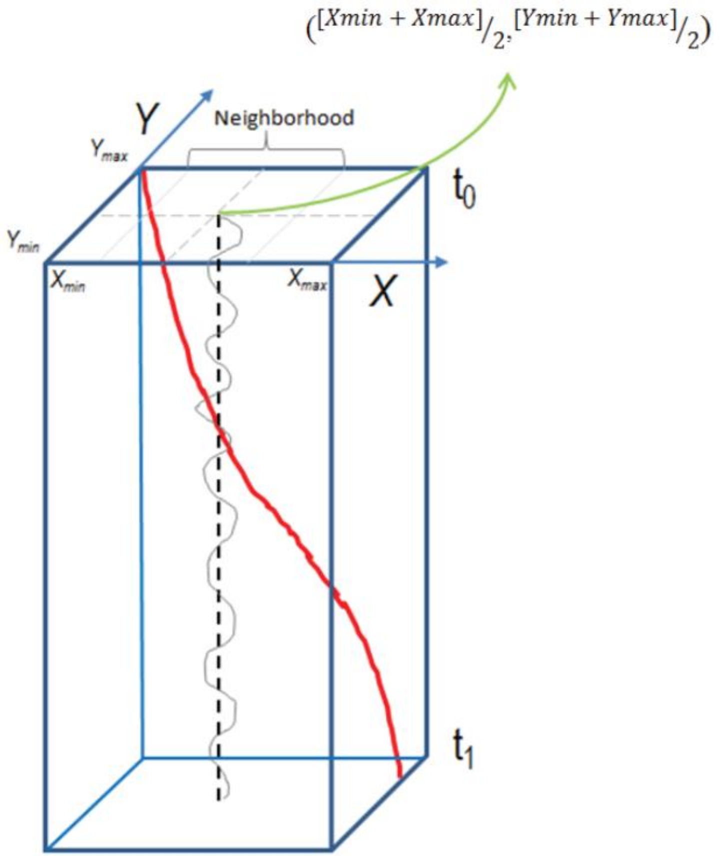
Statistical method

- It is possible to access statistic extraction even if no sonic log exists for the borehole.
- This method assumes that the embedded wavelet is the same as the truncated autocorrelation of the seismic trace.



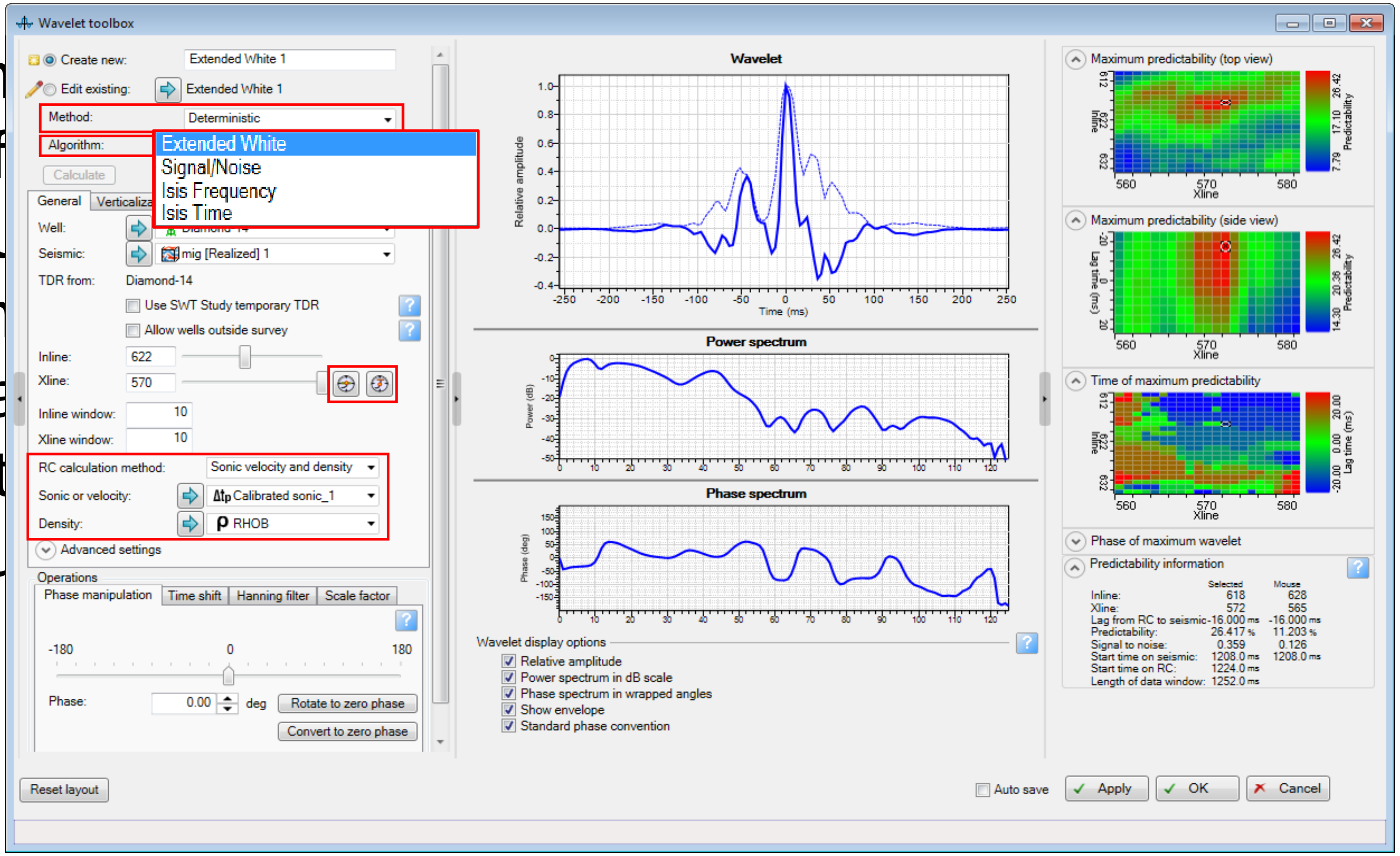
Principles of the Statistical method

- Given a borehole trajectory, this method extracts both and maximum values of X and Y: (Xmin, Ymin) and (Xm Ymax).
- The auto position (determined automatically) is calcul the mean between them:
    - *([Xmin+Xmax] /2, [Ymin+Ymax] /2)*
- It defines the position of the central trace for the extra
- Neighborhood (1x1, 3x3, 5x5) defines the number of tr used for the average.
- Unlike the Extraction algorithm, the Extended extractio has no limit on the number of traces used in the extrac window. It provides different options to limit the time window used in the extraction.



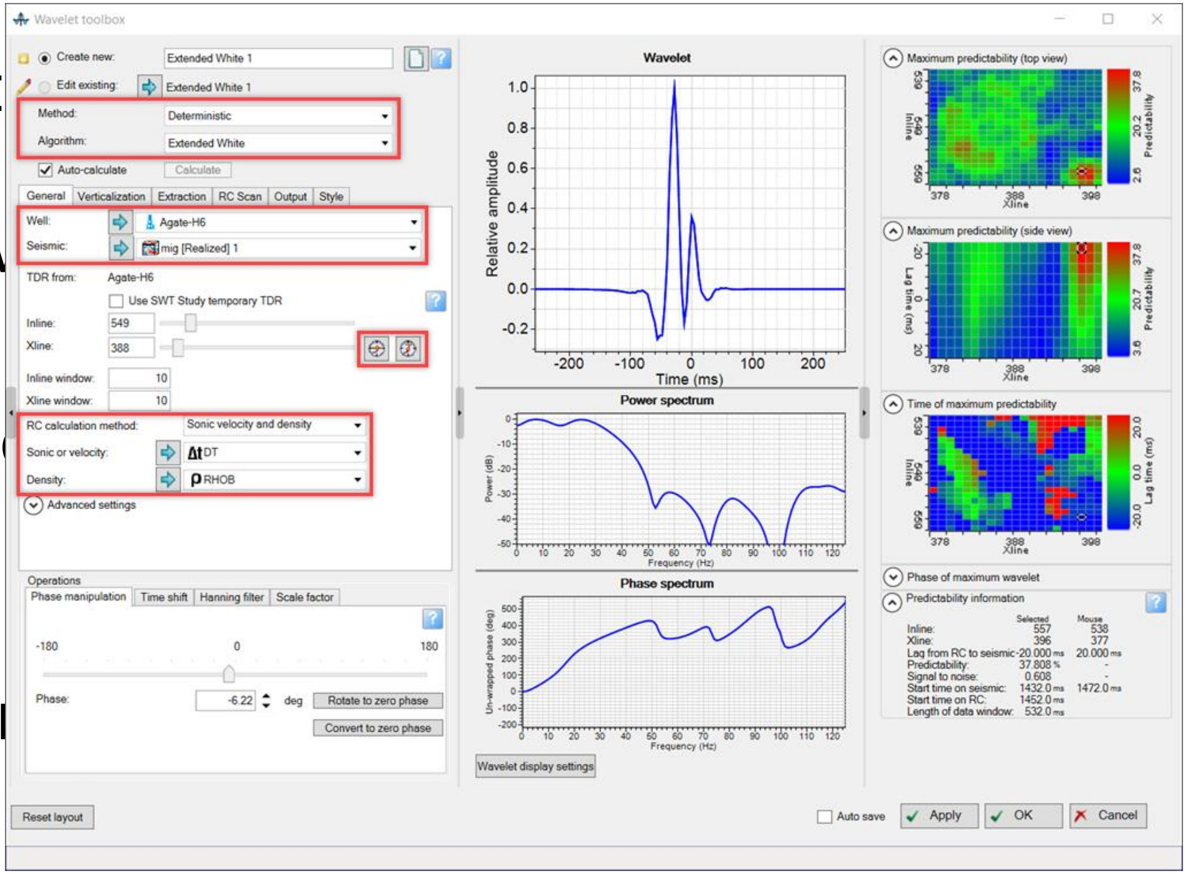
Deterministic method

- For this method, a seism volume and input logs of interest are required. You change the position of th extraction location intera based on predictability, t optimize the wavelet to u



Extended White

- The Extended Roy White wavelet extraction method decomposes tie and wavelet extraction into tw components:
  - Determination of optimal tim between well log reflectivity a seismic amplitude
  - Determination of the domina



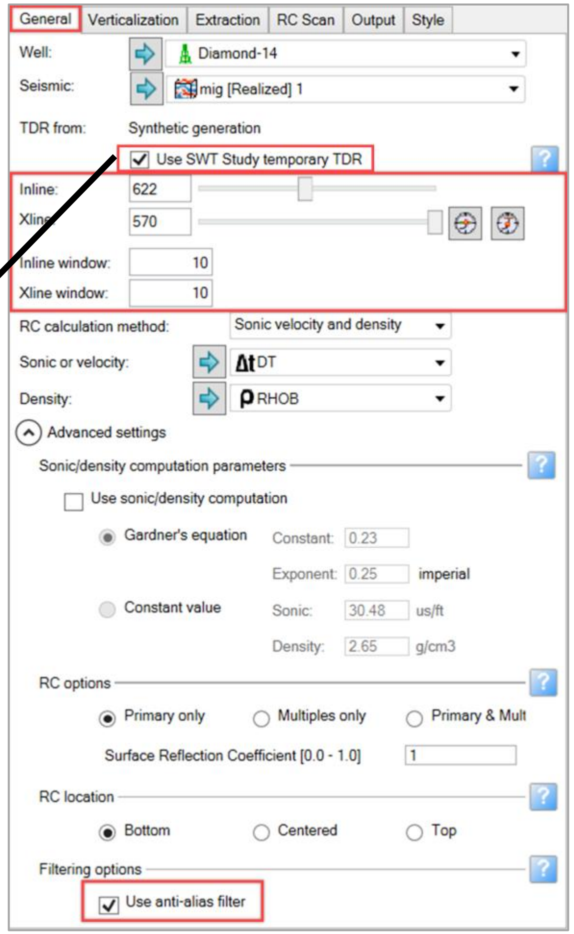
Settings for Extended White

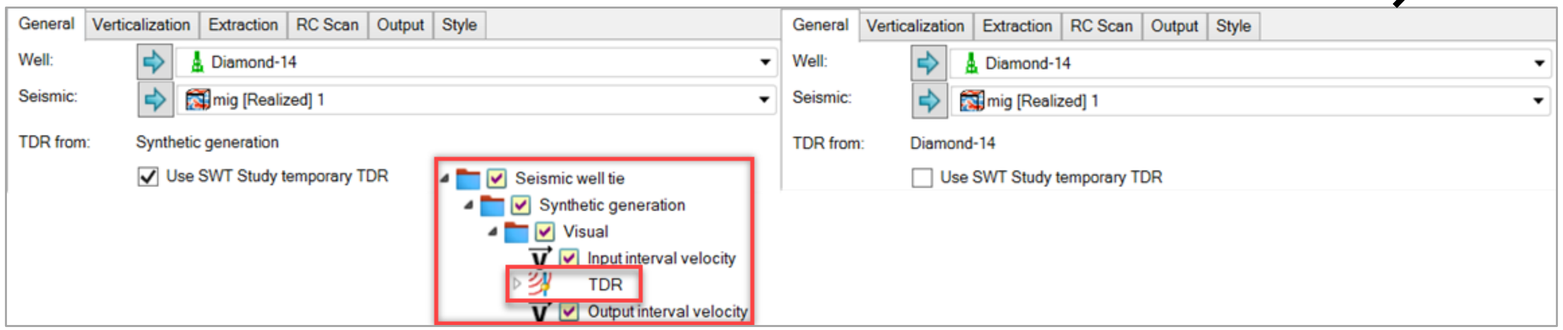
- After you select the deterministic method of wavelet extraction in the **Wavelet** toolbox, six tabs allow you to better define wavelet extraction.

  - General
  - Verticalization
  - Extraction
  - RC Scan
  - Output
  - Style

General tab

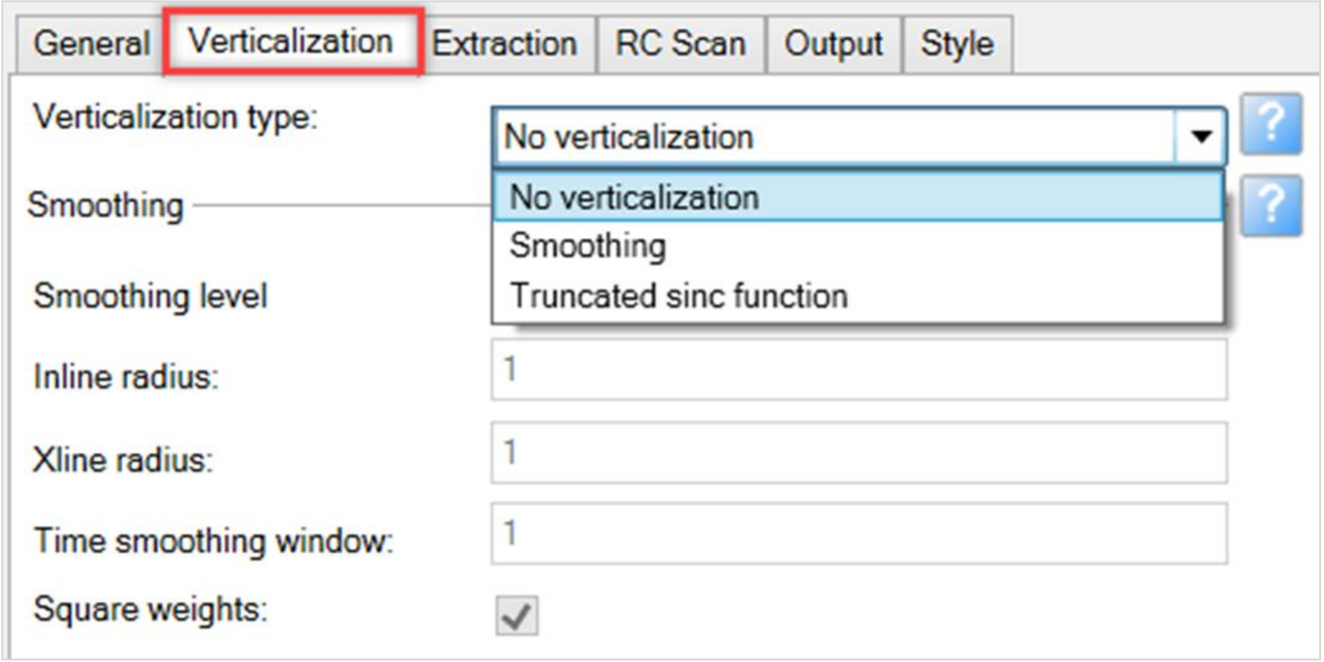
- In an active **Well section window**, you can use a temporary Time Depth Relationship (TDR) or the well TDR to extract a deterministic wavelet.
- If you use the SWT temporary TDR, any change from the bulk shift or stretch and squeeze process in the study is used by the **Wavelet** toolbox for the extraction.





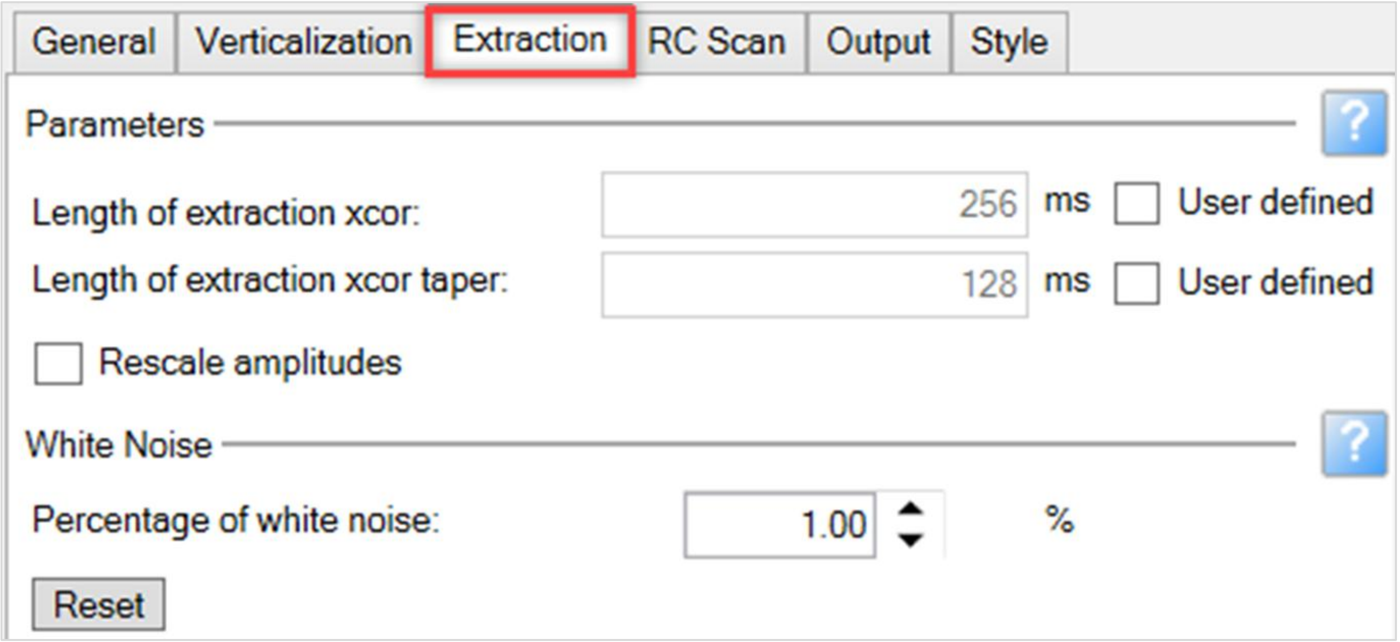
Verticalization tab

- A deviated trajectory often crosses several different seismic traces; the **Verticalization** tab offers options for merging them to extract a wavelet.



Extraction tab

- Select the *User defined* check boxes to customize the length of the taper applied to the cross-correlation that is computed and used for wavelet estimation.

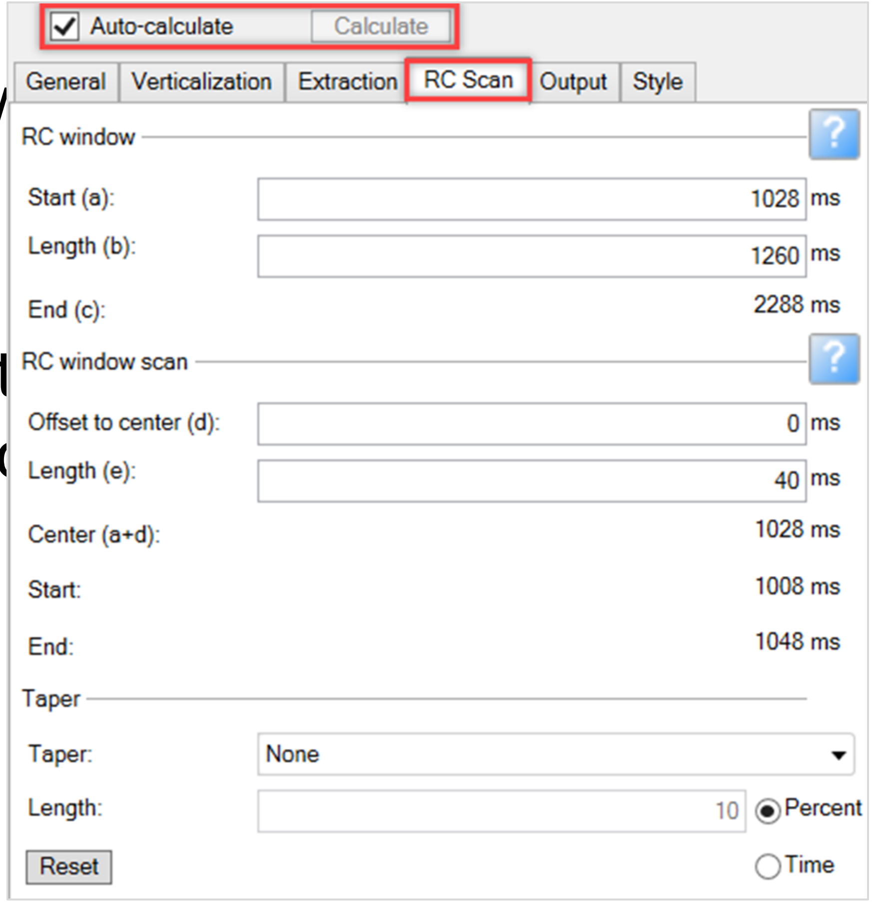


RC Scan tab

- Contains the options for setting the time v
the wavelet extraction process.

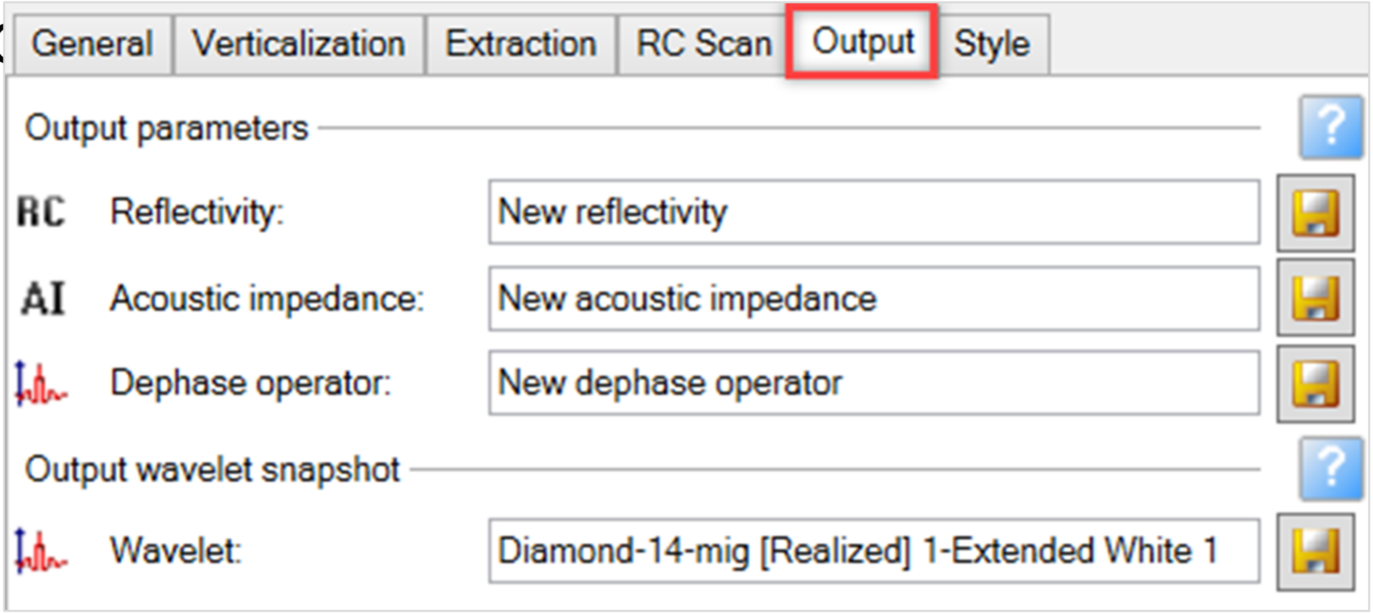
- Activate the *Auto-calculate* option to updat
wavelet automatically as parameters are c

- Alternatively, use the *Calculate* option afte
inserting the desired parameters.



Output tab

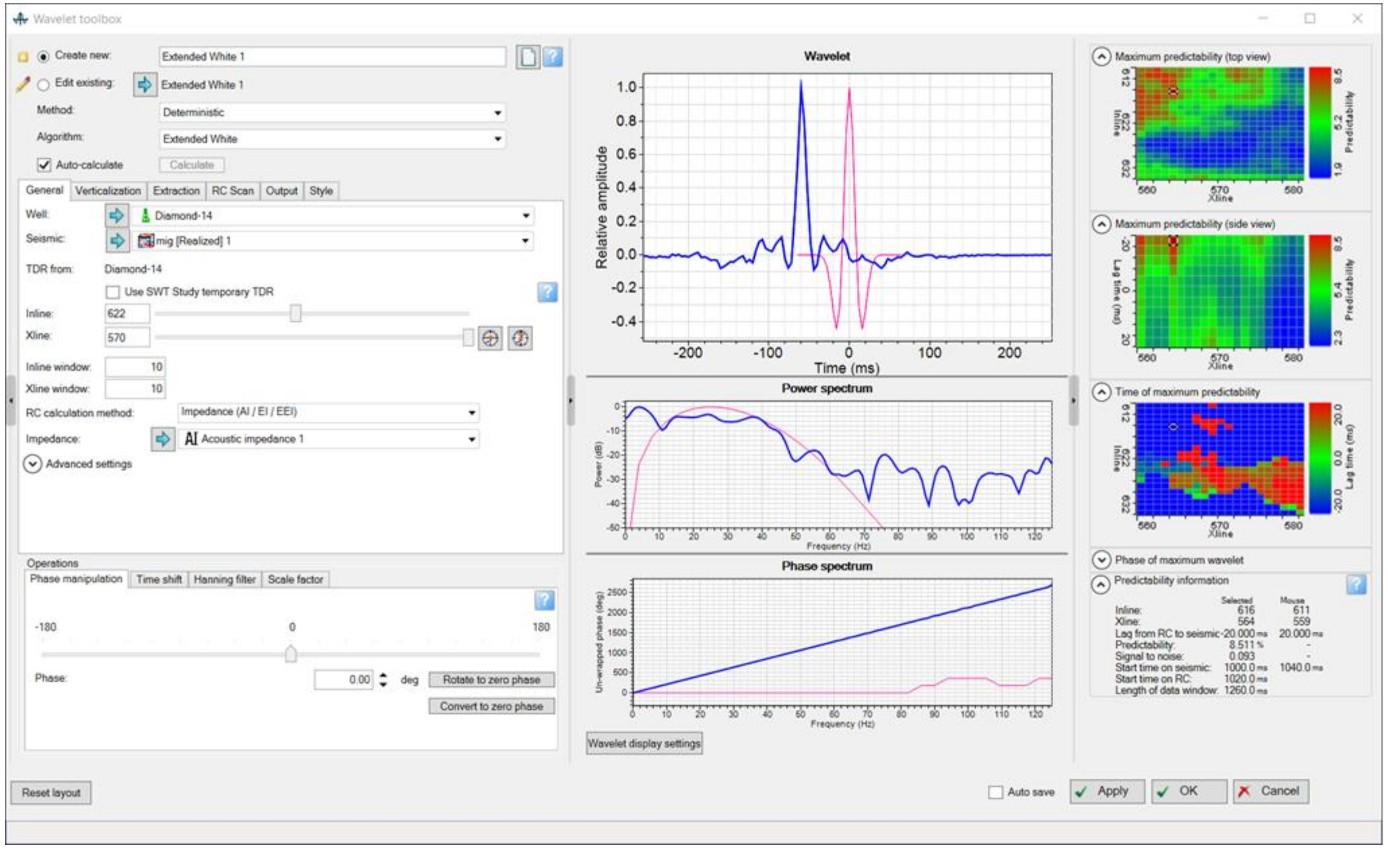
- Deterministic method can be used to generate Reflectivity, Acoustic Impedance, and the Dephase Operator as outputs. (Dephase Operator is a special purpose wavelet used t



- You also can save a wavelet snapshot.

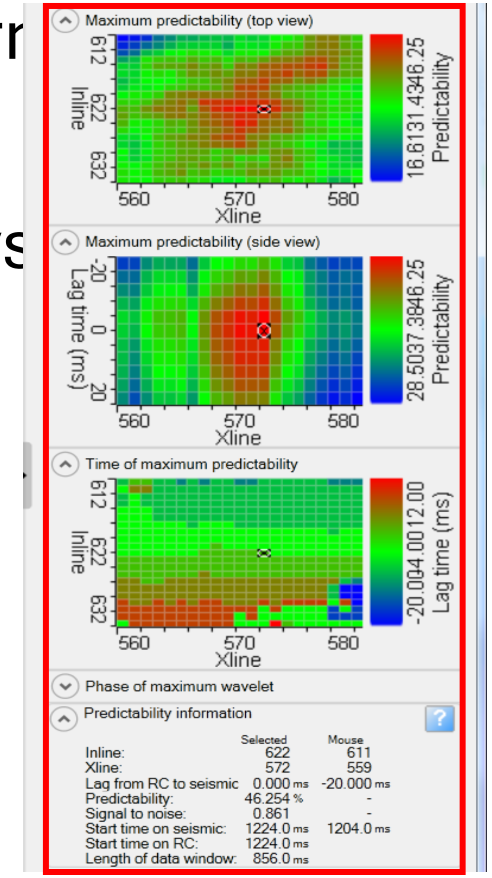
Wavelet extraction display (1)

- A **Predictability** section appears when you set all of the input parameters.
- Any change in the inputs or in the extraction parameters automatically updates the Predictability.



Wavelet extraction display (2)

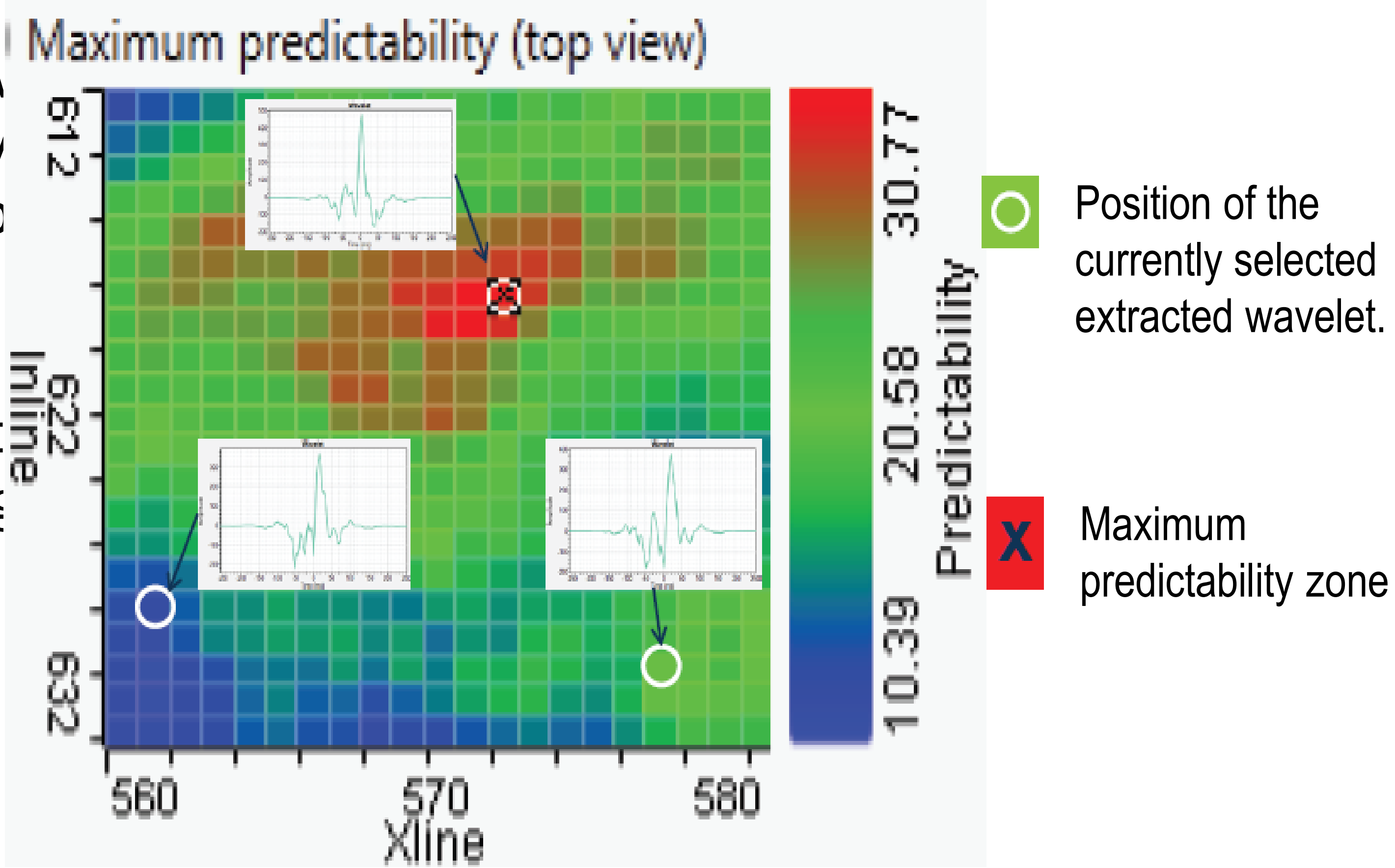
- Predictability in a Seismic well tie is used to deter
  optimal wavelet.
- The windows show different Predictability displays



- Maximum Predictability (top view)
- Maximum Predictability (side view)
- Time of maximum predictability
- Phase of maximum wavelet
- Predictability information

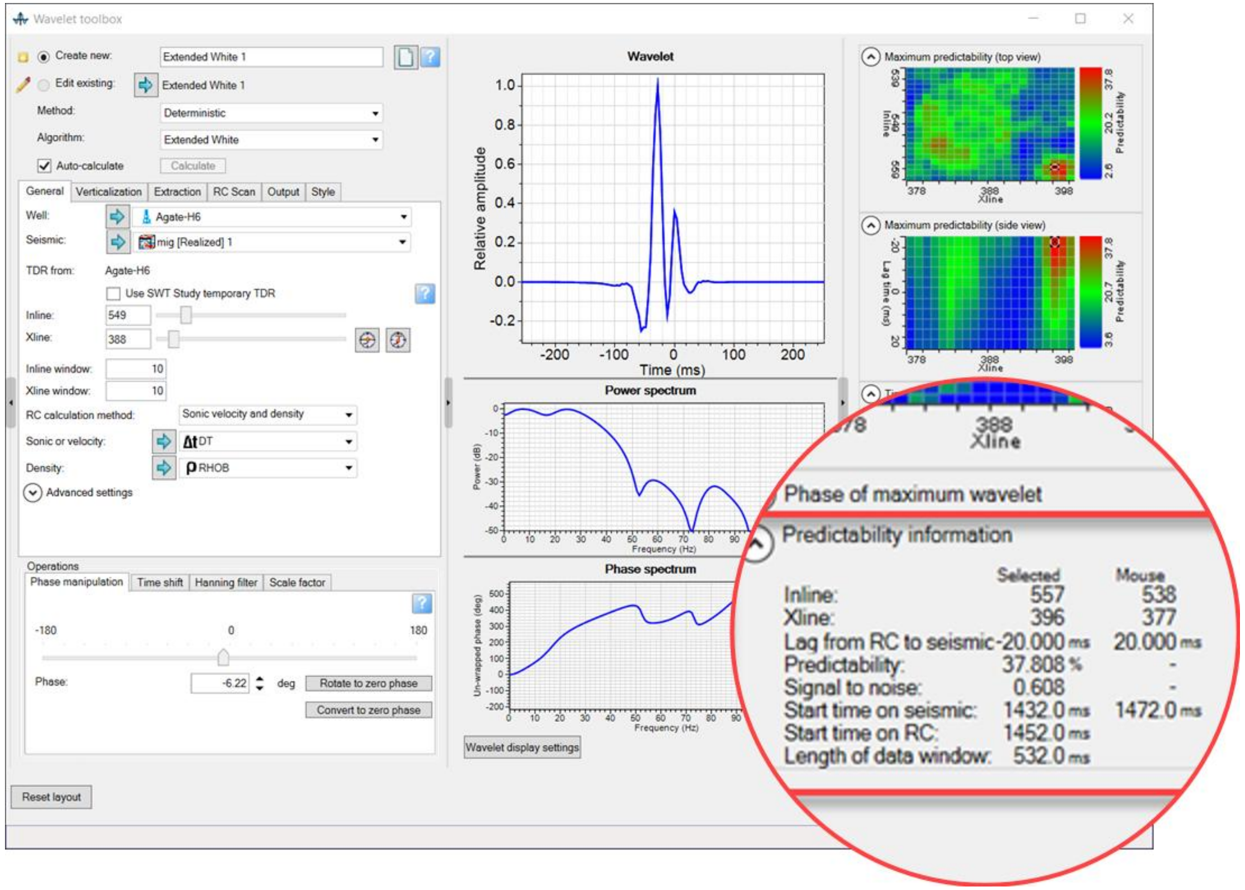
Wavelet extraction display (3)

- Initially, a white circle shov the currently selected wav at the maximum predictab of the cube of the extracte wavelet.
- The interpreter can move t extraction point by clicking any predictability display.
- A black **X** shows the maximum predictability zone.



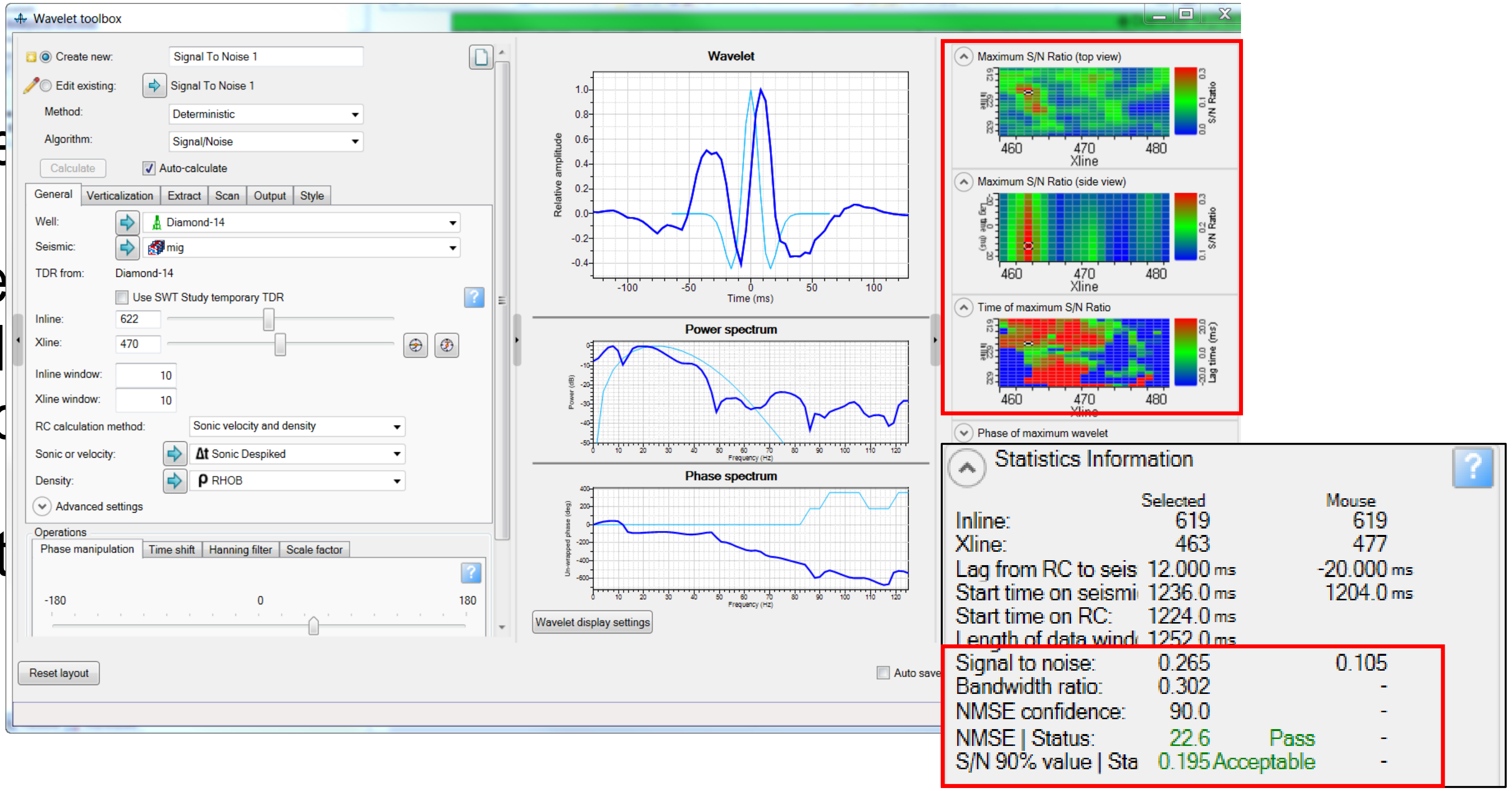
Predictability and correlation

- Predictability is a measure of the similarity of the underlying reflectivity.
- Simple cross-correlation methods do not benefit from these features.



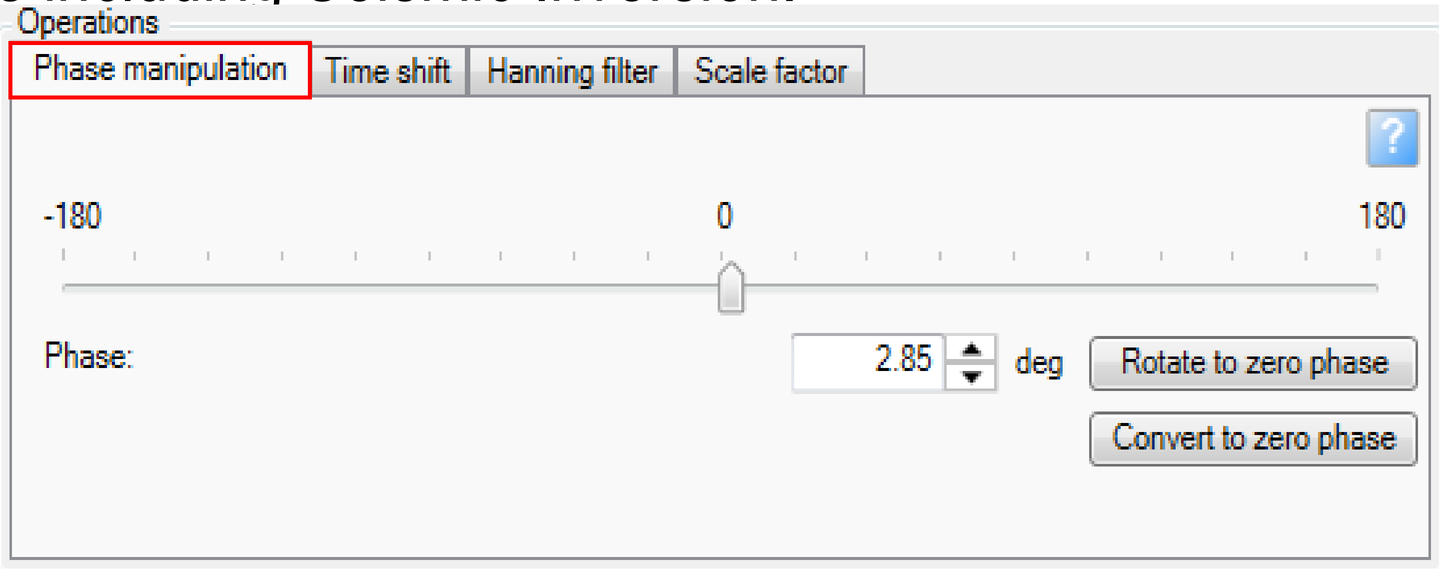
Signal/Noise methods for deterministic wavelets

- Based on Roy White, the Signal/Noise algorithm provide the best location based on Signal/Noise (S/N); that is, the ratio of the power in the signal (power in the smooth synthetic over the power in the residual the match with the seismic dat



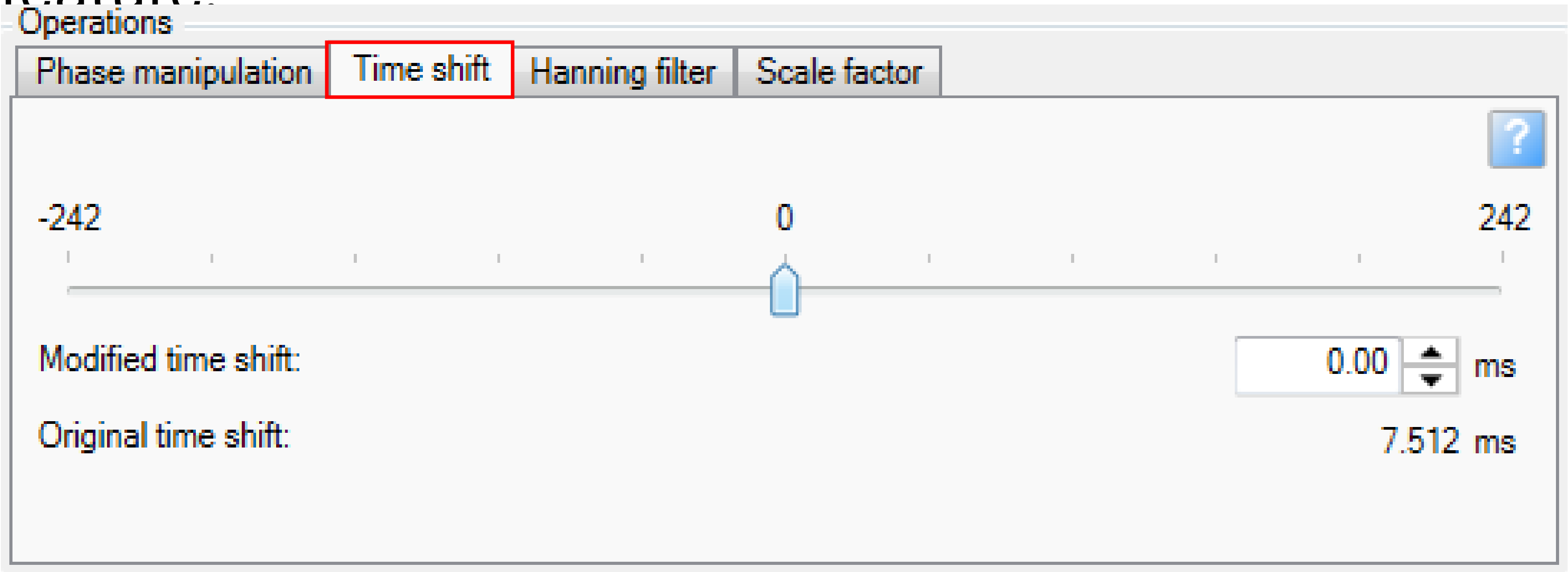
Phase manipulation

- Understanding the phase of the seismic data is important.
- Zero phase seismic data is essential for many geophysical analysis techniques including Seismic Inversion.



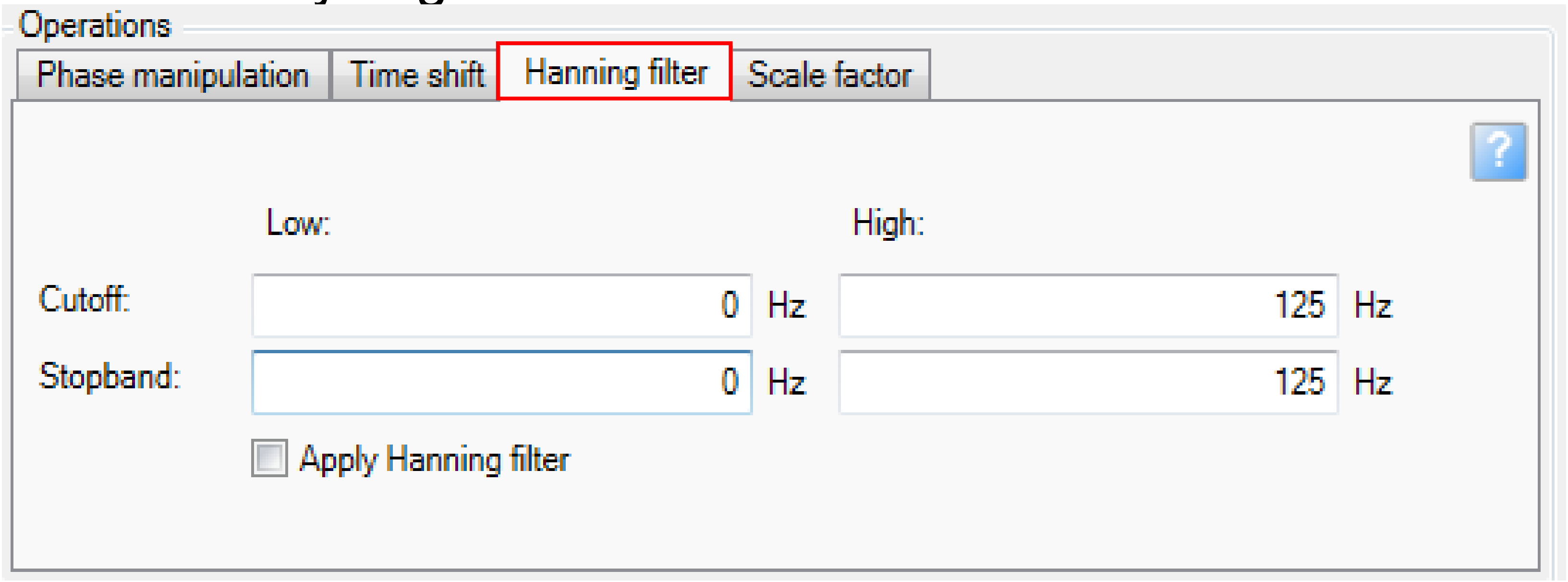
Time shift

- A bulk shift can be applied to the entire wavelet using the Time shift feature.



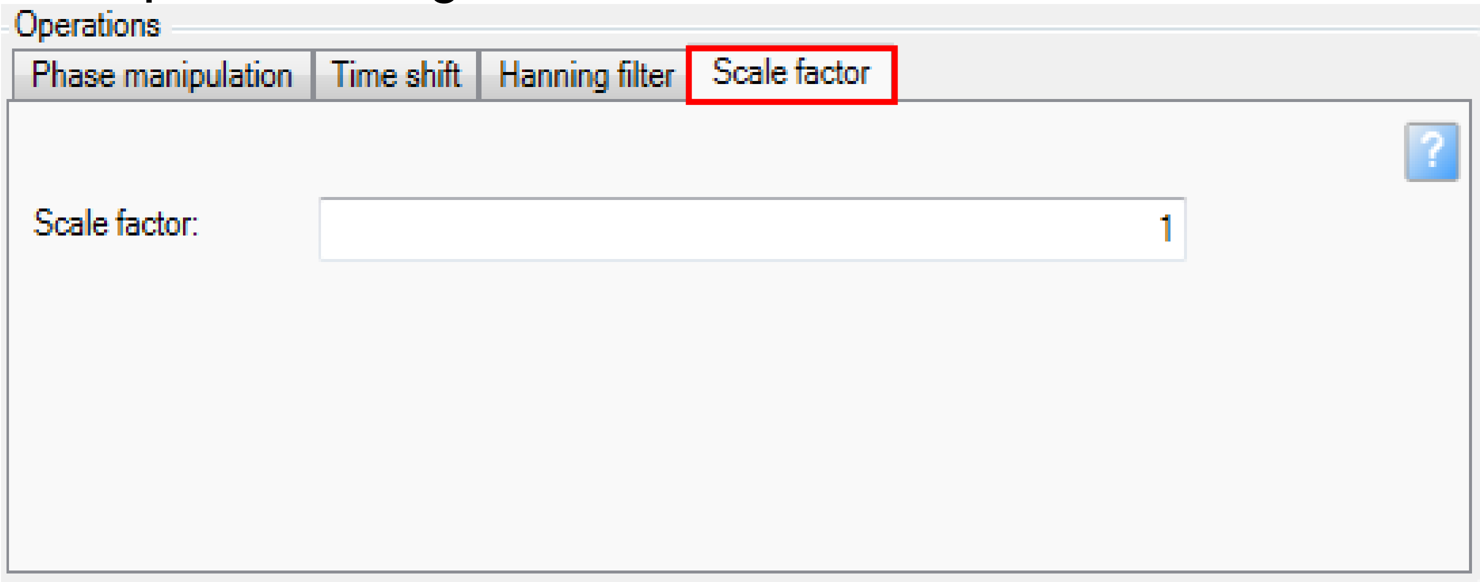
Hanning filter: Seismic data frequency characteristics

- It helps to evaluate if a wavelet extracted for a deeper interval is applicable for a shallower interval. This evaluation is done by filtering out the lower frequencies and analyzing tie at a shallow interval.



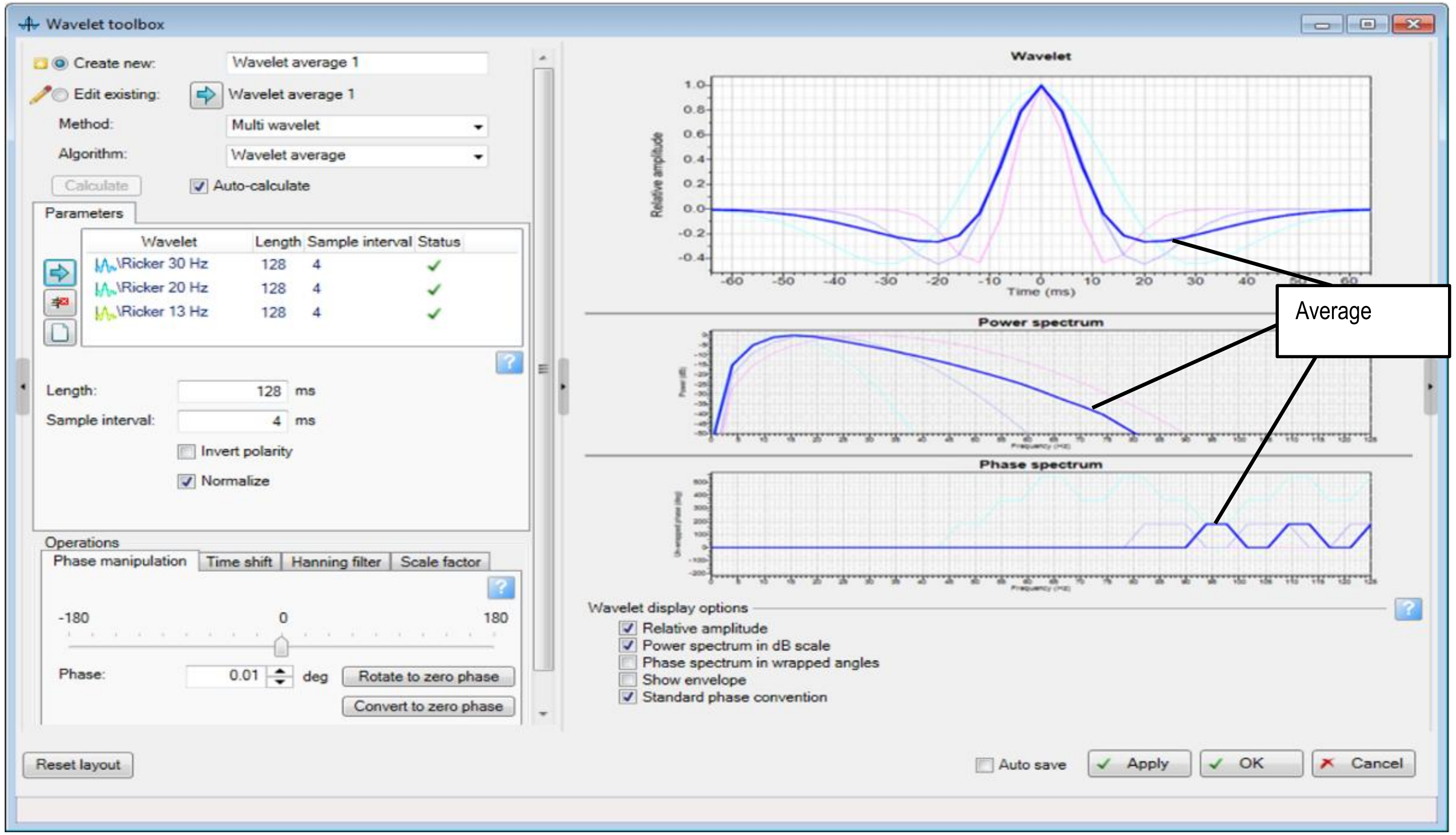
Scale factor

- Scales the amplitude of the wavelet.
- For the analytical and statistical normalized method of wavelet extraction in **Petrel**, the amplitude range varies from -1 to 1.



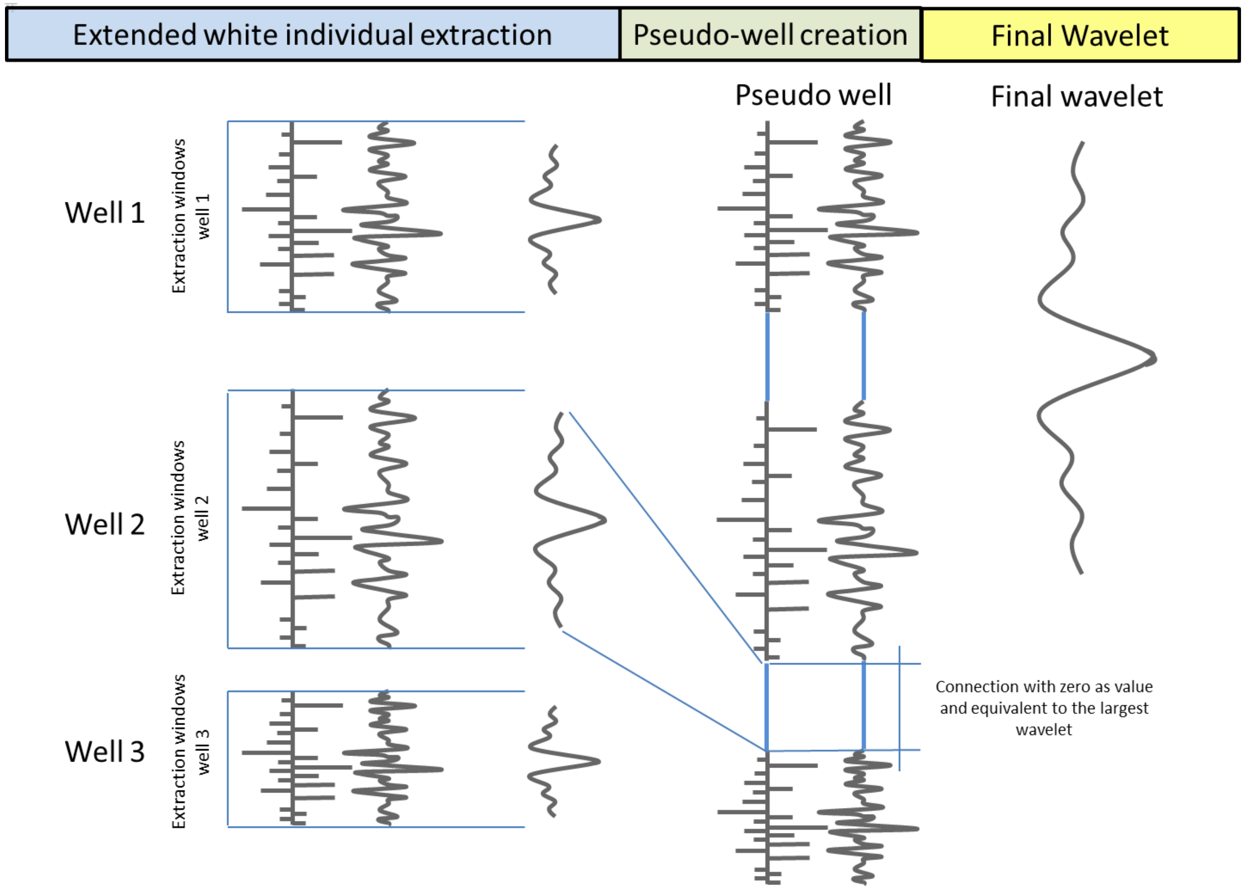
Multiwavelet

- Calculates an average wavelet from a set of selected wavelets.
- Defines the length and sample interval of the average wavelet.
- There is an additional option to invert the polarity of the resultant wavelet.



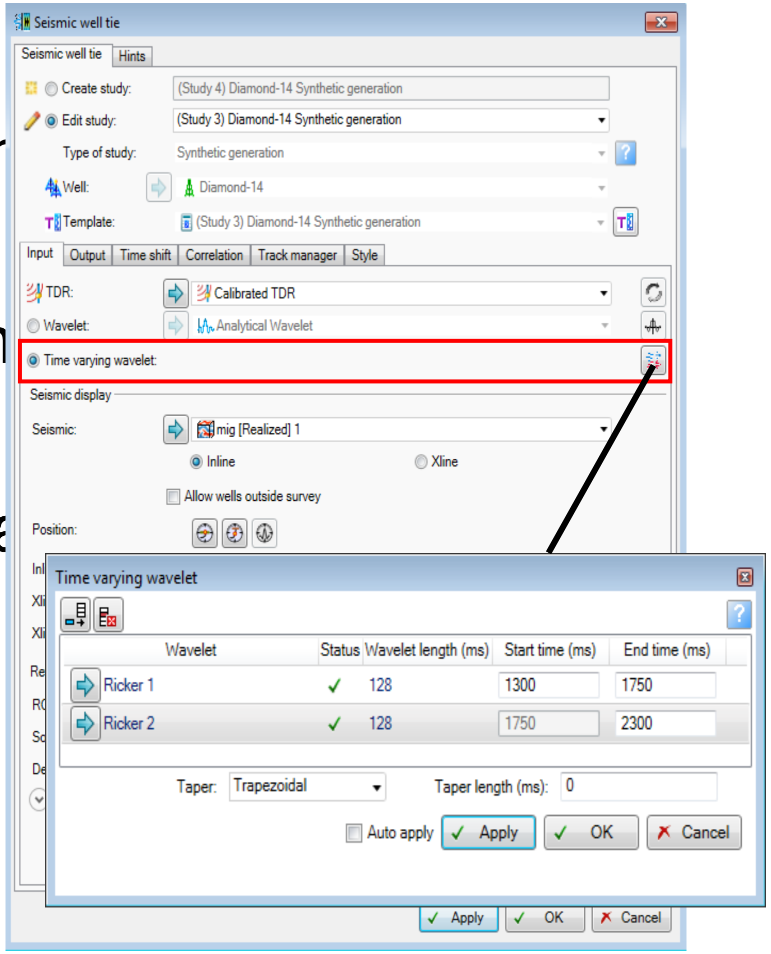
Multi-well Extended White wavelet extraction (MWEW)

- Extract a single wavelet from different wells.
- Use the extended Roy White method.
- Represent frequencies and amplitudes from different wells.



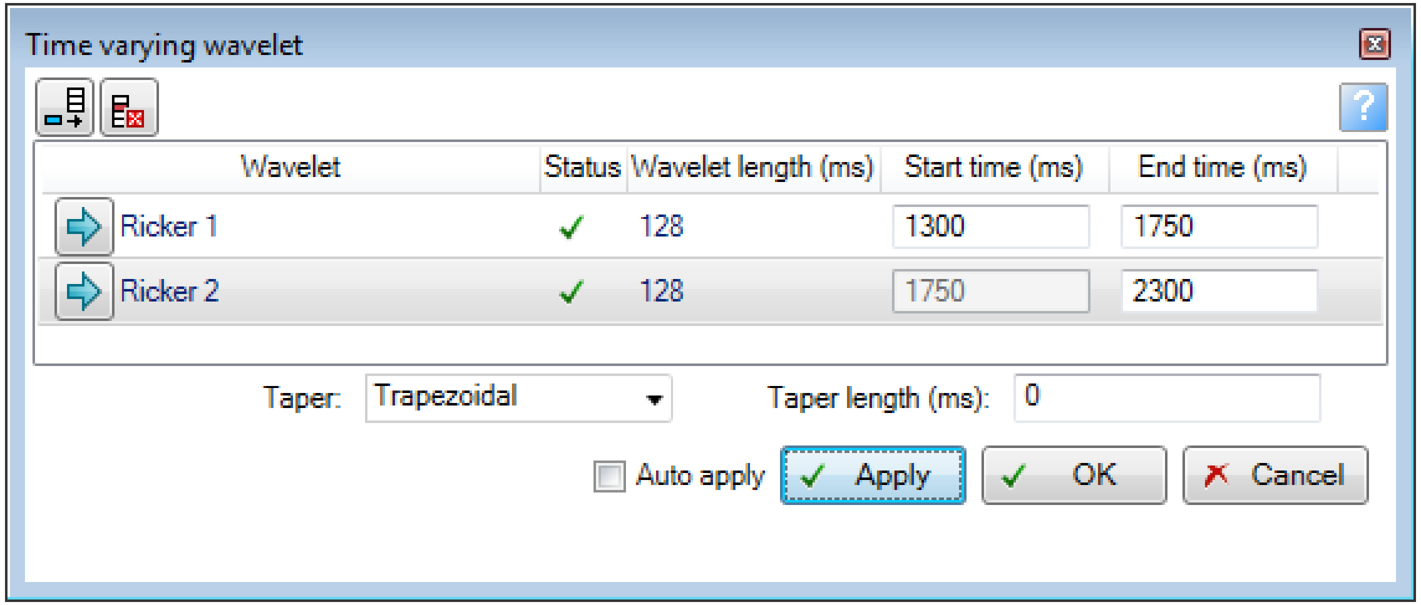
Time varying wavelet (1)

- To address decrease of frequencies and ar with depth.
- Applies existing wavelets over different tim to generate a synthetic.
- Each wavelet is used for a specified interva
- The minimum window that you can define



Time varying wavelet (2)

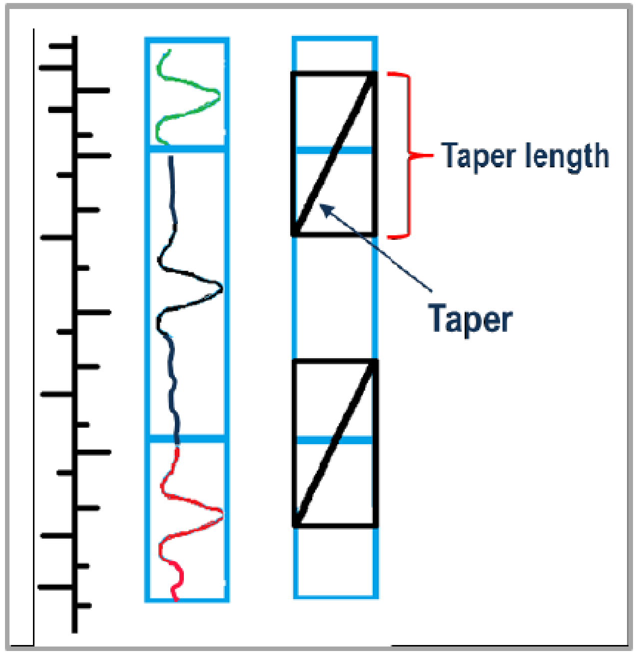
- Time ranges or time intervals can be specified over which a series of wavelets can be used to generate the synthetic.
- Create as many as ten time ranges represented in a dynamic table.

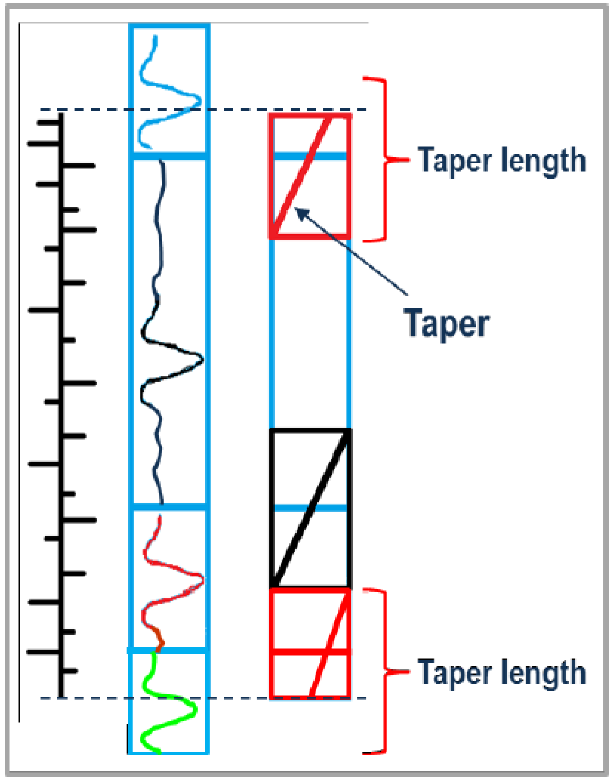


Time varying wavelet (3)

Taper length:

- A time in milliseconds over which the gate is gradually smoothed to its edges.
- This parameter is an integer value with maximum limits of the smallest time range.





Time varying wavelet (4)

- All the wavelets are resampled to the same sample rate of the one with the lowest sample interval.
- A synthetic seismogram is generated using the highest resolution.